AI – DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Cinnamon cultivation plays an important role in Sri Lanka's agricultural and export sectors; however, monitoring plant health, predicting growth, detecting diseases, and performing bark harvesting remain challenging due to environmental variability, the need for timely plant assessment, and the labor-intensive nature of harvesting. This research presents an AI-Driven Smart Cinnamon Growth Monitoring and Automated Harvesting System that integrates the Internet of Things (IoT), machine learning, computer vision, and robotic technology to support cinnamon cultivation and harvesting operations. The proposed system consists of three complementary components: an IoT-driven smart agriculture system for real-time cinnamon plant health monitoring, an AI-guided robotic machine for automated cinnamon bark harvesting, and a machine learning-based system for cinnamon growth prediction, disease detection, and robotic harvest guidance. The IoT component collects and presents relevant environmental and soil-related measurements, while the machine learning component analyses plant-related data and images to generate growth predictions, identify disease-related visual patterns, and provide harvesting-related guidance. The robotic component focuses on the mechanical and control requirements of cinnamon bark harvesting, creating a pathway for connecting analytical outputs with physical harvesting operations. The project follows a design, development, integration, and evaluation approach, incorporating hardware and software implementation, data processing, machine learning techniques, and system-level testing under defined conditions. The integrated prototype establishes a common technological framework for connecting agricultural monitoring, data-driven plant assessment, and robotic harvesting assistance. However, further validation is required to establish model reliability, sensor performance, harvesting accuracy, operational safety, and effectiveness under real plantation conditions. Overall, this research demonstrates a practical foundation for integrating intelligent technologies into cinnamon cultivation and identifies opportunities for further development towards reliable, data-driven, and sustainable agricultural practices in Sri Lanka.

Keywords: Smart Agriculture, Cinnamon Growth Prediction, Disease Detection, Internet of Things, Robotic Harvesting

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Computer Vision
DL	Deep Learning
IoT	Internet of Things
ML	Machine Learning
RGB	Red, Green, Blue
SDG	Sustainable Development Goal
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

1. INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

1.1.1 Background of Smart Agriculture

Agriculture supports food production, employment, economic development, and international trade. However, it faces challenges such as environmental changes, plant diseases, inefficient resource use, and labor-intensive activities. Smart agriculture uses technologies such as the Internet of Things (IoT), machine learning, computer vision, and robotics to improve agricultural monitoring, prediction, and operations [1]–[5].

IoT sensors collect environmental data, including temperature, humidity, and soil moisture. Machine learning identifies patterns and generates predictions, while computer vision analyses plant images. Integrating these technologies with robotics can connect monitoring and analysis with physical agricultural operations.

The proposed AI-Driven Smart Cinnamon Growth Monitoring and Automated Harvesting System consists of three components: IoT-based cinnamon plant health monitoring, AI-guided robotic cinnamon bark harvesting, and machine learning-based growth prediction, disease detection, and robotic harvest guidance.

1.1.2 Importance of Cinnamon Cultivation in Sri Lanka

Cinnamon, particularly Cinnamomum verum, is an important Sri Lankan agricultural and export crop. Its production involves cultivation, plant monitoring, harvesting, bark peeling, drying, and grading. Automation requires consideration of stem shape, bark characteristics, target positioning, and suitable harvesting methods.

Previous studies have investigated machine learning for cinnamon disease classification and species identification [6]–[8]. However, disease detection, growth prediction, and robotic guidance are different tasks. An integrated system must establish how analytical outputs support harvesting decisions and connect with the robotic mechanism.

1.1.3 Challenges in Cinnamon Plant Monitoring and Health Assessment

Temperature, humidity, and soil moisture vary over time and between cultivation areas. IoT monitoring provides repeated measurements to observe these changes [1], [2]. Plant symptoms can also be difficult to interpret because diseases, pests, nutrient deficiencies, and environmental stress may produce similar signs.

Image-based machine learning can support disease detection, but reliability depends on dataset quality, image conditions, and similarity between training and real-world environments [9], [10]. The IoT component monitors environmental conditions, while the machine learning component analyses plant-related data and images.

1.1.4 Role of IoT Sensors and Communication in Smart Agriculture

IoT performance depends on sensor selection, placement, sampling frequency, communication range, and power availability [1], [2]. The proposed monitoring component may measure temperature, relative humidity, and soil moisture. Readings should be time-to-timestamped and checked for missing or invalid values where supported.

Espressif Systems documents ESP-NOW, a connectionless communication protocol supported by ESP devices [11]. If used, its configuration should be described. Sensor errors, equipment faults, power interruptions, and communication failures may affect data quality. The IoT component provides environmental information for the wider system through the implemented interfaces.

1.1.5 Machine Learning for Cinnamon Growth Prediction

Growth prediction estimates plant characteristics such as height, stem dimensions, growth rate, or developmental stage. Machine learning can model relationships between growth outcomes and available inputs, including historical measurements and environmental data.

Previous research has explored machine learning for crop-yield prediction, plant growth prediction, and agricultural environments [12]–[15]. However, findings from other crops cannot automatically be applied to cinnamon.

The prediction target and period must be clearly defined. Predicting a future measurement differs from classifying growth stages or harvesting readiness. Regression models may be evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), while classification models may use precision, recall, and F1-score. Results must reflect the actual test data.

1.1.6 Image-Based Disease Detection and Multispectral Imaging

Image-based disease detection uses image processing and machine learning to identify patterns associated with plant health. Image classification assigns labels, object detection identifies locations, and segmentation identifies relevant image regions.

Research has demonstrated deep learning for plant disease detection [9], [10]. YOLO supports object detection, while MobileNets provides efficient neural-network architectures for vision applications [16], [17]. Transfer learning can adapt existing models to related tasks, although performance depends on dataset similarity and independent testing [18].

Multispectral imaging captures selected wavelength bands beyond conventional red, green, and blue (RGB) channels. Hyperspectral imaging generally captures many narrower spectral bands. Research has investigated spectral imaging for plant disease analysis and early virus detection [19]–[21].

The thesis must specify whether the implementation uses RGB, multispectral, hyperspectral, or combined data. Disease detection should be evaluated using labelled test data, with dataset limitations clearly documented. Literature coverage alone does not prove that multispectral imaging was implemented.

1.1.7 Computer Vision for Plant Analysis and Robotic Guidance

Computer vision supports image processing, feature analysis, recognition, and geometric reasoning [4]. Canny edge detection identifies boundaries based on intensity changes but cannot independently identify cinnamon plants or determine harvesting locations [5]. OpenCV provides image-processing tools, while visual-recognition benchmarks demonstrate the importance of defined datasets and evaluation procedures [22], [23].

In this project, computer vision may support disease analysis, plant-structure detection, and harvesting-target localization. Disease classification produces a category, whereas robotic guidance may require target coordinates or orientation.

Vision outputs must be translated into physical actions through appropriate calibration, coordinate transformations, and control mechanisms. Testing should establish whether the outputs can guide the harvesting mechanism under the evaluated conditions.

1.1.8 AI-Guided Robotic Cinnamon Bark Harvesting

The AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting investigates automation of bark removal. The workflow may include target identification, location estimation, mechanism positioning, harvesting, and outcome verification.

The actual level of automation—autonomous, assisted, operator-controlled, or simulated—must be clearly stated. Machine learning outputs may support harvesting decisions, but physical tool positioning may require additional sensing and control.

The mechanism must accommodate variations in stem dimensions, shape, and accessibility. Safety precautions, operator supervision, and emergency stopping should be documented where implemented.

Potential evaluation measures include target-localization accuracy, positioning error, task completion rate, harvesting time, repeatability, and plant damage. These should only be reported as results when supported by experimental measurements.

1.1.9 Integration of IoT Monitoring, Machine Learning, and Robotics

The three components have complementary responsibilities:

1. IoT monitoring: Collects and displays environmental and soil-related measurements.
2. Machine learning: Analyses relevant data and images to predict growth, detect disease-related patterns, and generate harvesting guidance.
3. Robotic harvesting: Uses suitable guidance and control inputs to perform harvesting operations.

Integration requires compatible data formats, communication mechanisms, clearly defined model outputs, and handling of missing data, communication failures, uncertainty, and unexpected machine conditions.

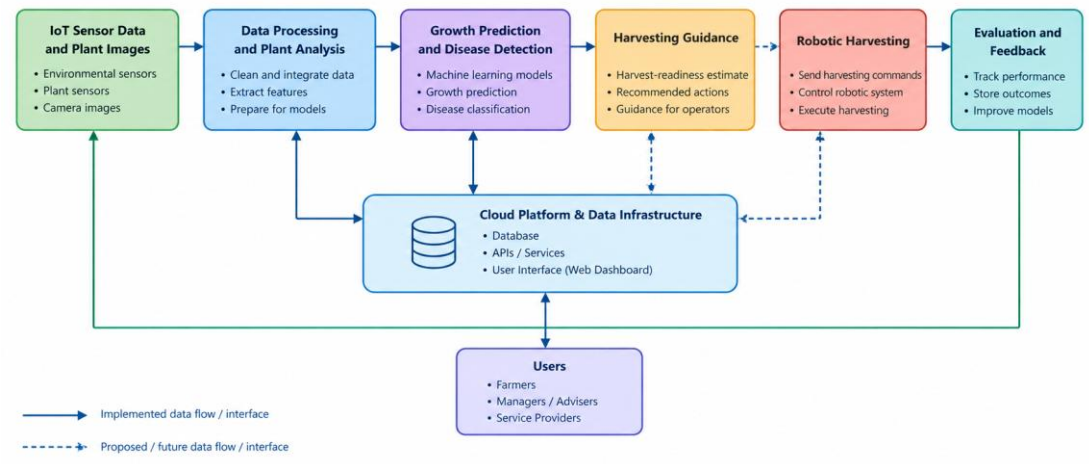


Figure 1 - overall system diagram

Having three functioning modules does not prove successful integration. Evidence must demonstrate correct data exchange and appropriate performance of the combined workflow. The project's contribution should therefore be assessed through implementation and testing rather than assumed benefits.

1.2 Stakeholders and User Needs

1.2.1 Stakeholder Identification

The main anticipated stakeholders are cinnamon growers, plantation managers, harvesting workers, machine operators, agricultural researchers, technical personnel, and the project team. Their needs inform system requirements. Unless interviews, surveys, or field observations were conducted, these needs should be described as anticipated rather than confirmed.

1.2.2 Cinnamon Growers and Plantation Managers

Growers and managers need clear information about cultivation conditions, plant health, growth, and harvesting. The monitoring interface may display sensor readings and identify missing data. Growth estimates and disease classifications may support decisions but should not be treated as definitive diagnoses or proof of harvesting readiness without validation.

1.2.3 Harvesting Workers and Machine Operators

Operators require safe machine operation, clear status feedback, and control over harvesting activities. Appropriate safety precautions and emergency stopping should be considered. The project investigates harvesting assistance and should not imply that human involvement is eliminated unless this is demonstrated.

1.2.4 Agricultural Specialists and Researchers

Researchers need traceable data, documented methods, and interpretable results. Disease models require details of image collection, dataset labels, and evaluation procedures. Growth models require clear prediction targets, input variables, and error measurements. Conclusions must remain within the conditions represented by the data.

1.2.5 Technical Personnel and System Maintainers

Technical personnel may maintain sensors, communication links, software, models, actuators, and mechanical components. Documentation should cover configuration,

calibration, software dependencies, model versions, troubleshooting, and component interfaces.

1.2.6 Summary of Stakeholder Needs

Table 1 - Stakeholder groups and anticipated needs

Stakeholder	Anticipated needs	Relevant component
Cinnamon growers	Environmental and plant information	IoT monitoring
Plantation managers	Information for cultivation and harvesting decisions	IoT and machine learning
Harvesting workers and operators	Safe and controlled operation	Robotic harvesting
Agricultural researchers	Traceable data and interpretable results	Monitoring and machine learning
Technical personnel	Maintainable hardware and documented interfaces	All components
Project team and supervisors	Clear responsibilities and evaluation evidence	Integrated system

These preliminary requirements should be refined using actual stakeholder feedback where available. The system should present monitoring information clearly, communicate analytical outputs transparently, and support controlled robotic operations.

1.3 Alignment with the Sustainable Development Goals

The project has potential relevance to SDG 9, SDG 12, and SDG 8. These represent intended contributions rather than confirmed impacts.

1.3.1 SDG 9: Industry, Innovation and Infrastructure

The project aligns with SDG 9 through the development and integration of IoT monitoring, machine learning, computer vision, and robotics [1]–[4], [16], [17]. Its contribution can be assessed through implemented functions, integration tests, and technical performance. A prototype alone does not establish commercial readiness.

1.3.2 SDG 12: Responsible Consumption and Production

Monitoring, growth prediction, disease detection, and harvesting guidance may support better operational decisions. However, reduced resource use, crop loss, or

material waste cannot be assumed. These benefits require comparative measurements of resource use, harvesting damage, or usable harvested material.

1.3.3 SDG 8: Decent Work and Economic Growth

The project explores agricultural automation and decision support in an industry that supports livelihoods in Sri Lanka. Its effects on productivity, employment, income, and working conditions depend on reliability, affordability, adoption, training, and maintenance. Cost analysis, practical testing, and stakeholder feedback are needed to assess these effects.

1.4 Purpose and Scope of the Common Report

1.4.1 Purpose of the Common Report

This report documents the shared problems, objectives, architecture, integration, implementation, and evaluation of the AI-Driven Smart Cinnamon Growth Monitoring and Automated Harvesting System. It explains how the three components relate and assesses their functionality, integration, and limitations. Individual theses provide detailed research and implementation evidence for each component.

1.4.2 Scope of Component 1: IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring

Component 1 collects and monitors selected environmental and soil-related measurements using the implemented sensors, controller, communication method, and interface. The common report describes its role in integration and data reliability, while detailed circuits, calibration, firmware, and component experiments belong in the individual thesis. If ESP-NOW is used, its configuration should be documented [11].

1.4.3 Scope of Component 2: AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting

Component 2 develops a robotic mechanism for cinnamon bark harvesting. Its scope may include the mechanical structure, harvesting tool, actuators, control electronics, positioning, and software. The actual level of automation must be demonstrated, and machine-learning guidance should be explained in terms of its interface and conversion into physical actions.

1.4.4 Scope of Component 3: Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance

Component 3 applies machine learning and image analysis to growth prediction, disease detection, and harvesting guidance. These tasks may use different inputs and outputs, including growth measurements, plant images, disease labels, and target-location information.

The report should document the actual datasets, models, imaging methods, and evaluation procedures. RGB, multispectral, and hyperspectral data must be distinguished, and only implemented methods should be claimed.

1.4.5 Scope of System Integration

System integration examines how environmental measurements and plant-related data are processed, how predictions or guidance are generated, and how output support harvesting. It requires defined data formats, interfaces, timing, and output interpretation.

Only connections and functions that were implemented or tested should be reported. Manual data transfers and simulated interfaces must be identified, and full end-to-end automation should not be claimed without a complete demonstration.

1.4.6 Research Boundaries and Limitations

The project evaluates a prototype and does not assume that all environmental conditions or plant diseases are monitored. Sensor reliability depends on calibration, placement, and hardware quality. Model performance depends on data, validation, and similarity between test and deployment conditions. Robotic performance depends on mechanical design, positioning, control, and plant variability.

Controlled tests may not represent all cinnamon-growing environments. Commercial deployment or improvements in yield, export quality, labour productivity, cost, or sustainability cannot be claimed without suitable evidence. These limitations define what the project can demonstrate and what requires further work.

1.4.7 Relationship Between the Common Report and Individual Theses

The common report presents the shared context, objectives, architecture, component relationships, integration strategy, and group-level evaluation. Each individual thesis documents its component's literature, requirements, technical design, implementation, experiments, and results.

The common report explains how individual contributions support the overall system without repeating every technical detail. Individual and shared responsibilities must

be acknowledged transparently, and terminology, diagrams, objectives, and implementation claims should remain consistent across all reports.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

Cinnamon cultivation involves interconnected activities, including monitoring growing conditions, identifying plant health problems, understanding growth, determining harvesting suitability, and performing bark harvesting. These activities require different information and skills. Environmental monitoring provides growing-condition measurements, image analysis identifies visual symptoms, growth prediction estimates plant development, and harvesting requires accurate target identification and physical handling of suitable stems.

Although digital technologies support agricultural monitoring and analysis, applying them to cinnamon presents specific challenges. IoT systems collect environmental measurements, while machine learning analyses agricultural data and images. However, these technologies do not automatically ensure that information can be combined effectively or used to guide physical harvesting. Each function has distinct data requirements, processing methods, and performance considerations.

Ray [1] and Alahmad et al. [2] discuss IoT and sensor-based information in smart agriculture. Mohanty et al. [9] and Neik et al. [10] examine machine learning for image-based plant disease detection, while Van Klompenburg et al. [12] review machine learning approaches to crop yield prediction and Nesteruk et al. [13] investigate plant growth prediction through intelligent embedded sensing. These studies provide relevant technological foundations, although their findings cannot automatically be generalized to cinnamon-specific applications.

Cinnamon presents additional challenges because its commercially valuable product is bark. Harvesting requires consideration of stem characteristics and bark-removal requirements. Unlike image classification or numerical prediction, robotic harvesting must translate information into controlled physical actions through appropriate target identification, positioning, mechanical control, and outcome evaluation.

The central problem addressed by this group thesis is the challenge of developing and integrating reliable environmental monitoring, machine learning-based cinnamon growth and disease analysis, and AI-guided robotic bark harvesting into a coordinated system that can be evaluated under defined operating conditions.

The problem concerns not simply the availability of IoT devices, machine learning algorithms, or robotic machinery, but their adaptation to cinnamon cultivation, the

usefulness of their outputs for related tasks, and the demonstration of individual and integrated performance through testing.

The project addresses this problem through three complementary components:

1. IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring: Collects and presents relevant environmental and soil-related measurements.
2. AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting: Addresses the mechanical and control requirements of harvesting.
3. Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance: Analyses relevant data and images to generate growth predictions, disease classifications, and harvesting-related guidance.

The monitoring component provides environmental information, the machine learning component generates analytical outputs, and the robotic component provides physical harvesting capability. Where supported by the implemented architecture, these functions can be connected to establishing a workflow from data collection and analysis to harvesting guidance and execution.

The overall problem will be addressed to the extent that the required component functions are demonstrated, intended interfaces are established, and the integrated workflow is evaluated against defined criteria. Improvements in yield, labor costs, or harvesting efficiency are not assumed and require experimental evidence.

2.2 Root Causes and Overall Research Gap

2.2.1 Fragmented Monitoring and Agricultural Decision-Making

Environmental observations, plant health assessments, growth information, and harvesting activities may be handled separately. Sensors monitor environmental conditions, visual inspection or image analysis assesses plant health, and harvesting requires additional information about plant characteristics and the harvesting method.

When these activities operate independently, information from one may not be available to another. Environmental measurements may support growth analysis, but their usefulness depends on consistent recording and integration with appropriate growth data. Similarly, disease classification does not automatically provide the target location or physical characteristics required for robotic harvesting.

IoT-based systems improve environmental data availability [1], [2], but reliable measurements and their relevance to intended decisions remain essential. Data

collection alone does not provide a complete plant health assessment or automated harvesting capability.

The project addresses this challenge through defined monitoring, analytical, and robotic components while investigating how their outputs can be coordinated. The research focuses on establishing appropriate interfaces and evaluating the resulting workflow.

2.2.2 Challenges in Environmental Monitoring

Environmental monitoring depends on the consistency and availability of sensor measurements. Calibration errors, unsuitable placement, hardware faults, power interruptions, and communication failures can affect readings. Measurements from different locations or intervals may also be difficult to compare without appropriate timestamps and source identification.

IoT technologies support agricultural data collection and transmission [1], [2]. However, the implementation must define monitored variables, sampling frequency, transmission methods, and procedures for handling missing or invalid data.

The cinnamon monitoring component must establish reliable measurement acquisition and make the selected data available to users or relevant software. If these data support machine learning, their format, timing, and relationship to the prediction task must also be defined.

The research gap is therefore not the general absence of agricultural sensors, but the need to develop and evaluate a monitoring arrangement suitable for cinnamon cultivation and determine whether its measurements are reliable for the intended application.

2.2.3 Challenges in Growth Prediction and Disease Detection

Growth prediction and disease detection involve different analytical tasks. Growth prediction estimates a defined characteristic or outcome, whereas disease detection classifies evidence associated with plant health. Both depend on appropriate input data, model selection, and evaluation.

Van Klompenburg et al. [12] review machine learning for crop yield prediction, while Nesteruk et al. [13] investigate plant growth prediction through intelligent embedded sensing. These studies provide methodological foundations, but crop yield prediction, general plant growth prediction, and cinnamon growth prediction are distinct tasks. Cinnamon-specific models require clearly defined targets and relevant data.

Disease detection presents additional challenges. Mohanty et al. [9] demonstrate deep learning for image-based plant disease detection, while Neik et al. [10] review machine learning approaches to crop disease detection. These studies demonstrate the potential of image analysis and the importance of suitable training data and evaluation.

For cinnamon, performance may be affected by lighting, backgrounds, image quality, growth stages, symptom appearance, and disease-category distribution. A model performing well on a limited dataset may not achieve equivalent results under different field conditions.

Furthermore, disease detection and robotic guidance require different outputs. A disease classifier may produce a category and confidence score, whereas robotic harvesting may require target coordinates and stem dimensions. The system must define how analytical outputs are interpreted and what additional information is needed before a physical action.

The third component addresses the development and evaluation of machine learning functions for cinnamon-related tasks and investigates how their outputs can support harvesting guidance. These capabilities must be demonstrated using the implemented data and evaluation procedures.

2.2.4 Challenges in Automated Cinnamon Bark Harvesting

Robotic harvesting introduces physical and control challenges beyond monitoring and prediction. The mechanism must reach the target, position its tool, execute the required movement, and operate within the plant's physical constraints and the machine's design limitations.

Cinnamon bark harvesting requires consideration of stem geometry and bark removal. Variations in stem dimensions, alignment, accessibility, and surface conditions may affect harvesting performance.

AI-based guidance can assist target identification or selection, but analytical predictions do not guarantee successful physical execution. Relevant information must be translated into commands that the robotic mechanism can interpret. This depends on the sensing arrangement, target localization, coordinate interpretation, mechanical structure, and control system.

Evaluation should reflect the harvesting task and may include target localization error, positioning accuracy, task completion rate, cycle time, repeatability, and observed plant or bark damage.

The research gap concerns developing and testing a harvesting mechanism suited to the defined cinnamon bark harvesting task and determining whether it can operate using the guidance available from the implemented system.

2.2.5 Integration and Interoperability Challenges

The three components must operate as a coordinated system rather than unrelated prototypes. Individual modules may function correctly but encounter difficulties when connected.

Potential issues include incompatible data formats, inconsistent timing, missing measurements, communication delays, coordinate-system differences, and uncertainty in machine learning outputs. These challenges become particularly important when analytical information guides physical actions.

For example, a model may identify a target in an image, but the robotic controller must interpret its location for positioning. Similarly, IoT measurements cannot automatically support growth prediction unless the model's input requirements and data-processing procedures accommodate them.The project must specify exchanged data, producing and receiving components, and procedures for unavailable or unsuitable information. Manual data transfers and operator intervention must be documented rather than presented as fully automated integration.

The overarching research gap is the need to investigate the coordinated application of cinnamon-focused monitoring, machine learning-based plant analysis, and robotic bark harvesting within a defined, testable system. The project addresses this through component development, interface design, integration, and performance evaluation.

2.2.6 Summary of Root Causes and Research Gap

Table 2 - Root causes, resulting challenges, and project responses

Root cause or challenge	Resulting problem	Relevant project component
Environmental measurements may be incomplete, inconsistent, or difficult to access	Limited availability of reliable monitoring information	IoT monitoring
Plant growth and disease analysis require appropriate data and validated models	Uncertain growth estimates or disease classifications	Machine learning

Harvesting requires target identification, positioning, and physical control	Analytical information may not translate into a successful harvesting operation	Robotic harvesting
Components may use incompatible inputs, outputs, or interfaces	Independent modules may not operate as a coordinated system	All three components
Performance may vary across data and operating conditions	Results may not generalize beyond the tested setup	All three components

The overall research gap concerns both component-level performance and system-level integration. The project must determine whether each component performs its intended function and whether the implemented interfaces support common cinnamon monitoring and harvesting workflow.

Evaluation will rely on prototype implementation and testing evidence. Conclusions about practical deployment, productivity improvements, or economic and environmental benefits must remain consistent with the evidence collected.

2.3 Subproblem and Component Mapping

The common problem is divided into three interdependent subproblems corresponding to the project components. Each has distinct technical responsibilities, inputs, outputs, and evaluation requirements. Their interdependence arises from their potential contribution to a shared workflow, not from an assumption that every component directly exchanges data with every other component.

2.3.1 Subproblem 1: Environmental Monitoring and Data Availability

Corresponding component: IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring.

The first subproblem concerns collecting and presenting reliable environmental and soil-related measurements relevant to cinnamon cultivation. Without appropriate monitoring, users may lack consistent access to the selected environmental variables.

The component investigates how sensors, processing, and communication mechanisms can collect measurements and present them through the monitoring interface. Its design must address sensor selection, data acquisition, communication reliability, measurement intervals, and missing or invalid readings.

Expected outputs include recorded sensor measurements and an accessible representation of monitored conditions. Historical data, where retained, may support

analysis of changes over time. Their suitability for machine learning depends on the implemented interface and model requirements.

Evaluation may include measurement error against an appropriate reference, successful transmission rate, missing-data rate, update interval, and monitoring availability. Selected metrics must reflect the actual hardware and experimental setup.

This subproblem establishes the system's environmental data collection capability. However, environmental measurements alone cannot determine disease status or confirm harvesting suitability.

2.3.2 Subproblem 2: Automated Physical Cinnamon Bark Harvesting

Corresponding component: AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting.

The second subproblem concerns executing cinnamon bark harvesting through a robotic mechanism. The challenge is to design and control a mechanism capable of performing the required operation under the conditions supported by the prototype.

The component may involve mechanical design, tool selection, actuator control, target positioning, and movement coordination. Where AI-generated guidance is used, the mechanism must interpret it and translate it into suitable physical commands.

Inputs depend on the implemented design and may include operator commands, target coordinates, positioning measurements, or guidance from the machine learning component. Outputs may include completed harvesting actions and observations or measurements describing their outcomes.

Evaluation may examine task completion, positioning accuracy, cycle time, repeatability, and observed bark or plant damage. The team must define how each measure is recorded and avoid claiming unmeasured benefits.

This subproblem provides the physical capability to execute and evaluate harvesting operations, which monitoring and machine learning cannot perform independently.

2.3.3 Subproblem 3: Machine Learning-Based Growth Prediction, Disease Detection, and Harvest Guidance

Corresponding component: Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance.

The third subproblem concerns analyzing cinnamon-related data to generate growth predictions, disease classifications, and harvesting guidance. These tasks require suitable data, defined outputs, appropriate models, and systematic evaluation.

Growth prediction requires a defined target, such as a measurable growth characteristic or developmental outcome. Disease detection requires labelled examples of the categories to be distinguished. Harvest guidance requires a clear definition of the information needed for target selection or harvesting decisions.

Depending on the architecture, the component may use plant images, historical measurements, environmental data, or combinations of these sources. Each function's input must be documented separately rather than assuming all models use identical data.

Numerical growth predictions can be evaluated using appropriate regression metrics. Disease detection can use precision, recall, F1-score, and a confusion matrix. If object detection supports harvesting guidance, evaluation may include detection precision, recall, and localization accuracy.

The outputs may support robotic harvesting when the necessary data and control interfaces are implemented. However, growth predictions and disease classifications are not sufficient instructions for physical movement unless their relationship to the harvesting action is explicitly established.

This subproblem provides the analytical capabilities that distinguish the proposed system from monitoring alone, investigating how data and image analysis can support plant assessment and harvesting guidance.

2.3.4 Interdependence of the Three Subproblems

The three subproblems contribute to a common monitoring and harvesting workflow. The IoT component provides environmental measurements, the machine learning component analyses relevant data and images, and the robotic component performs physical harvesting.

Figure 2.1: Relationship between the three project components

Environmental and soil measurements → Plant monitoring and data collection

Plant images and relevant measurements → Growth prediction and disease detection

Suitable analytical outputs and target information → Robotic harvest guidance and physical harvesting

The arrows represent potential information flows and functional dependencies. They should appear as direct system connections in the final architecture only where those connections have been implemented.

Table 3 - Mapping of the common problem to the three subproblems

Component	Main subproblem	Primary output	Evaluation focus
IoT-driven monitoring	Collecting and presenting reliable environmental measurements	Sensor readings and monitoring information	Measurement accuracy, data availability, transmission reliability
Robotic harvesting	Executing controlled physical harvesting operations	Robotic movements and harvesting outcomes	Task completion, positioning, cycle time, repeatability
Machine learning	Predicting growth, detecting disease, and generating harvesting guidance	Predictions, classifications, and guidance outputs	Prediction error, classification performance, localization where applicable

Together, the three components address the common problem from complementary perspectives. The monitoring component establishes access to selected environmental data, the machine learning component interprets relevant information, and the robotic component provides physical harvesting capability. System-level testing determines whether their implemented functions can be coordinated and whether the overall workflow meets its intended requirements.

2.4 Overall Aim and Group Objectives

2.4.1 Overall Aim

The overall aim of the project is:

To design, develop, integrate, and evaluate an IoT-driven cinnamon plant monitoring and machine learning-guided robotic harvesting system that combines environmental data collection, cinnamon growth prediction and disease detection, and automated bark harvesting under defined experimental conditions.

This aim unifies the three components while emphasizing implementation and evaluation rather than assuming benefits before experimental validation.

2.4.2 Group Objectives

The project will pursue the following objectives.

Objective 1: Establish the overall system requirements and architecture.

Define the common problem, component responsibilities, hardware and software requirements, interfaces, and intended information flow.

Evaluation evidence: Requirements, architecture diagrams, interface definitions, and traceability between requirements and implemented functions.

Objective 2: Develop and evaluate the IoT-based monitoring component.

Implement the selected sensors and communication mechanisms to collect and present the environmental or soil-related measurements supported by the prototype.

Evaluation evidence: Sensor measurements, reference comparisons where available, communication tests, missing-data observations, and monitoring update performance.

Objective 3: Develop and evaluate the robotic cinnamon bark harvesting component.

Implement the mechanical and control functions required for harvesting and evaluate performance under controlled conditions.

Evaluation evidence: Operational demonstrations, task completion records, positioning measurements where applicable, cycle times, and harvesting quality or damage observations.

Objective 4: Develop and evaluate the machine learning component.

Implement and test the defined growth prediction, disease detection, and harvesting guidance functions using appropriate datasets and evaluation procedures.

Evaluation evidence: Dataset descriptions, model configurations, independent test results, appropriate prediction or classification metrics, and localization measurements where object detection is implemented.

Objective 5: Integrate the implemented components through defined interfaces.

Connect relevant component inputs and outputs were supported by the architecture and verify correct information exchange and interpretation.

Evaluation evidence: Interface tests, data exchange records, integration test cases, observed errors, and demonstrations of the connected workflow.

Objective 6: Evaluate the integrated system and identify its limitations.

Assess system performance under tested conditions and document technical limitations, dependencies, and opportunities for further development.

Evaluation evidence: End-to-end test results where the complete workflow is implemented, acceptance-criteria comparisons, failure observations, and documented limitations.

2.4.3 Objective-to-Component Mapping

Table 4 - Group objectives and evaluation evidence

Objective	Primary responsibility	Evaluation evidence
Objective 1: Define requirements and architecture	Shared group responsibility	Requirements, architecture, and interface documentation
Objective 2: Develop IoT monitoring	Component 1	Measurement and communication tests
Objective 3: Develop robotic harvesting	Component 2	Harvesting task and mechanical performance tests
Objective 4: Develop machine learning	Component 3	Prediction, classification, and guidance evaluation
Objective 5: Integrate components	Shared group responsibility	Interface and integration tests
Objective 6: Evaluate the complete system	Shared group responsibility	System-level test results and limitations

The objectives are assessed through defined evidence rather than unsupported numerical targets. Before testing, suitable acceptance criteria should be established based on the hardware, model requirements, project constraints, and experimental design.

For example, monitoring measurements may be compared with a reference instrument, machine learning may be assessed using a separate test dataset, and robotic harvesting

may be evaluated through repeated trials. Integration testing must establish whether connected components exchange and interpret information correctly.

The overall aim will be considered achieved to the extent that the implemented functions are demonstrated, their performance is evaluated appropriately, and the system's integration level and limitations are documented.

2.5 Scope, Assumptions, and Constraints

2.5.1 Project Scope

The project covers the design, development, integration, and evaluation of three related components for cinnamon cultivation and bark harvesting.

The IoT component covers the implemented monitoring variables, sensors, communication mechanisms, and software functions. The machine learning component addresses defined growth prediction, disease detection, and harvesting guidance tasks. The robotic component covers the mechanical and control functions required for the harvesting operation supported by the prototype.

The common report addresses the shared problem, system architecture, integration strategy, group objectives, and combined evaluation, while detailed designs and experiments are documented in the individual theses.

Evaluation is limited to the available hardware, software, datasets, and operating conditions. Simulated, manually performed, or future-development functions must be identified clearly.

2.5.2 Assumptions

The project is developed under the following assumptions, which must be revised if inconsistent with the actual implementation.

1. **Availability of relevant plant material or data:** Suitable cinnamon plants, images, measurements, or other data will be available for development and evaluation.
2. **Availability of sensor hardware:** Selected sensors and processing devices can be configured to collect the required environmental measurements.
3. **Availability of suitable image data:** Relevant images or labelled observations will be available for growth, disease, or harvesting guidance tasks.
4. **Controlled testing conditions:** Initial testing can be conducted under conditions that allow performance observation and measurement.

5. **Defined component interfaces:** Required data and control interfaces can be specified within hardware and software limitations.
6. **Operator supervision where required:** Robotic testing can be conducted with appropriate supervision and safety precautions.

These are planning assumptions, not confirmed facts. Their validity should be reviewed during implementation and documented.

2.5.3 Technical Constraints

The project may be constrained by sensor accuracy, processing capacity, communication reliability, power availability, and environmental conditions affecting monitoring hardware.

Machine learning may be limited by dataset size, class imbalance, image quality, labelling consistency, computing resources, and development time. Predictions may not generalise when training data inadequately represent the plants and conditions encountered during testing.

Robotic performance may be constrained by mechanical structure, actuator capability, positioning accuracy, tool geometry, control response, and variations in cinnamon stems.

Integration introduces additional constraints when components use different communication mechanisms, processing platforms, and data formats. These differences must be addressed through interface design and testing.

Specific constraints and their effects should be established through implementation and experimental evidence rather than assumed in advance.

2.5.4 Operational Constraints

Testing may involve limited numbers of plants, images, sensor configurations, and harvesting trials. Consequently, results may not represent performance across all cinnamon-growing environments.

Environmental variation, lighting, plant growth stage, stem geometry, and target accessibility can affect performance. Uncontrolled factors should be recorded as part of the experimental conditions.

Robotic testing requires appropriate operating procedures to minimise injury and unintended plant damage. Manual positioning, supervision, or operator confirmation requirements must be stated explicitly.

Available time, funding, equipment, and technical resources may also determine which functions can be implemented and which remain future extensions.

2.5.5 Data and Model Constraints

Machine learning performance depends on the information represented in the training and testing data. A dataset containing only selected disease categories cannot establish the model's ability to recognise other diseases. Similarly, growth prediction cannot be assumed to support variables or prediction periods absent from its training and evaluation.

Data leakage between training and testing sets must be avoided. Evaluation should use appropriate data separation and task-specific metrics, with class imbalance, missing measurements, preprocessing assumptions, and known limitations documented.

Environmental measurements must also be distinguished from direct evidence of plant health. Sensor readings describe growing conditions but do not independently establish a disease diagnosis.

Harvesting guidance is constrained by its available information. A growth estimate or disease classification does not automatically establish harvesting readiness; any relationship between model outputs and harvesting decisions must be defined and evaluated.

2.5.6 Integration Dependencies

System integration depends on the correct operation of individual components and their interfaces. Sensor data must match the receiving software's expected format, machine learning outputs must be interpretable by receiving components, and robotic commands must correspond to supported mechanical operations.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

The proposed project combines IoT-driven cinnamon plant monitoring, AI-guided robotic bark harvesting, and machine learning-based growth prediction, disease detection, and harvest guidance within an integrated agricultural support system. Rather than treating monitoring, analysis, and harvesting independently, the system connects them through shared data and coordinated decision-making.

Traditional cinnamon cultivation and harvesting rely heavily on manual observation and experience, which can limit continuous monitoring and consistent assessment. The proposed solution complements growers' knowledge with sensor measurements, data-driven analysis, and robotic assistance.

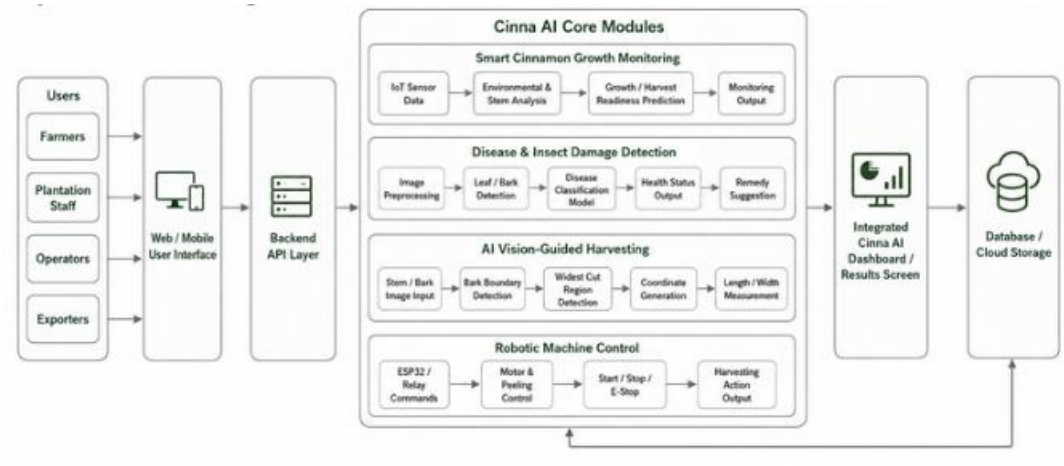


Figure 2- system overview

The integrated solution comprises three components.

Component 1: IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring. This component collects environmental and plant-related measurements using selected IoT sensors, potentially including temperature, humidity, and soil moisture. The measurements provide updated information about cultivation conditions for monitoring and further analysis.

Component 2: AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting. This component performs harvesting-related mechanical operations using a robotic mechanism and control inputs. Its design considers stem characteristics, positioning, movement, operational safety, and minimizing plant damage.

Component 3: Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance. This component analyses available sensor data, images, and other relevant observations to estimate growth-related outcomes, identify potential disease symptoms, and generate harvesting guidance.

The components follow a shared workflow: Component 1 supplies measurements, Component 3 analyses available data and generates predictions or guidance, and Component 2 executes supported harvesting operations. Robot status may also be communicated to users where the necessary interfaces are implemented.

A shared user interface and data-management layer are proposed to present measurements, analytical results, harvesting recommendations, and robot status while organizing exchanged information and historical records. These are architectural elements whose implementation depends on the selected technologies and project constraints.

Overall, the system aims to improve access to plant information, support informed decisions, and reduce dependence on entirely manual assessment and harvesting. Its effectiveness must be established through individual component testing and integrated system evaluation.

3.2 High-Level Architecture

The proposed architecture contains five logical layers: **user interaction, IoT monitoring, data and communication, machine-learning and decision support, and robotic harvesting**. These layers define system responsibilities and information flows rather than a fixed hardware or software platform.

User interaction layer: Enables growers or authorized operators to review monitoring data, analytical results, and harvesting status. Where implemented, users may configure monitoring parameters, provide feedback, or approve harvesting operations. Approval and emergency stopping are important operational controls for physical robotics.

IoT monitoring layer: Contains sensors and embedded devices that collect cultivation measurements and transmit them through the selected communication interface. Preliminary validation and organization by plant, location, or timestamp may also be included.

Data and communication layer: Exchanges information between components through defined protocols, formats, storage, and communication services.

Measurements should include timestamps and plant or location identifiers where available to maintain traceability between observations and analytical results.

Machine-learning and decision-support layer: Processes relevant sensor data, historical observations, and available plant images to generate growth predictions, potential disease classifications or alerts, and harvesting guidance. The models, features, confidence thresholds, and decision rules are defined in Component 3's individual thesis. Outputs must be communicated consistently to users and, where appropriate, the robot.

Robotic harvesting layer: Contains the robotic mechanism and control functions that execute supported harvesting instructions within defined operating limits. Operational feedback should be communicated to users where the required interfaces exist. Safety controls must remain effective if the network, monitoring system, or machine-learning service becomes unavailable.

The architecture distinguishes **analytical guidance from physical control**. A growth prediction or disease assessment does not independently confirm harvesting readiness. Instructions must satisfy the project's decision rules, operating constraints, and safety checks before execution.

The accompanying architecture diagram represents this conceptual design and should be updated to reflect the devices, protocols, deployment arrangements, and interfaces actually implemented.

3.3 Component 1 Summary: IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring

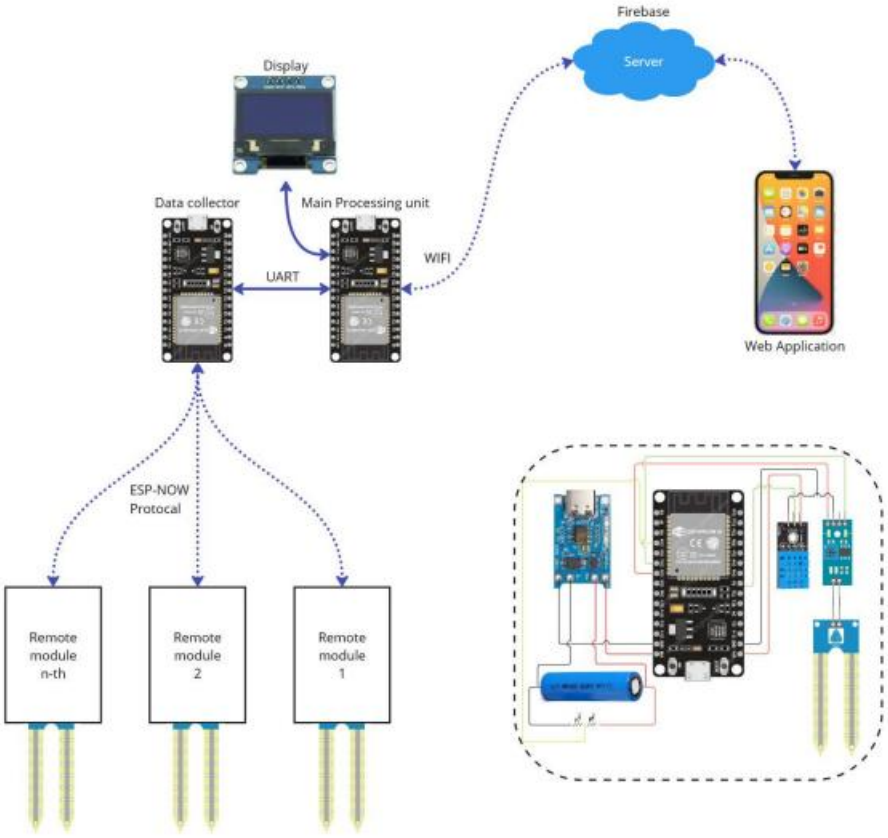


Figure 3 - component 1 diagram

Subproblem addressed: Limited access to continuous, structured, and timely information about cinnamon cultivation conditions and plant-health-related parameters.

Inputs: Measurements from selected environmental or plant-related sensors, device status, and available timestamps or identifiers.

Core method: IoT sensors and embedded devices collect cultivation measurements and transmit them to the monitoring or data-management service. Depending on the implementation, functions may include sensor interfacing, periodic acquisition, basic validation, wireless communication, and measurement display.

Outputs: Current or recent readings, monitoring summaries, historical records where storage is implemented, and alerts where suitable thresholds are defined. These outputs support growers and provide relevant inputs for Component 3.

Contribution to the overall solution: Establishes environmental monitoring and supplies structured sensor data for analysis. Integration with Component 3 allows measured conditions to be considered alongside other available information when assessing growth or plant health.

Main interfaces: Sensor-to-device connections, device-to-data-service communication, monitoring displays, and data exchange with Component 3. Any interface with Component 2 depends on the final integration design.

Evaluation focus: Sensor accuracy against suitable reference instruments, transmission reliability, update frequency, data completeness, and correct display of readings. Disease detection and growth prediction must be evaluated separately under Component 3 rather than inferred from sensor performance.

Detailed sensor selection, circuit design, embedded implementation, communication methods, and experimental results are documented in Component 1's individual thesis.

3.4 Component 2 Summary: AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting

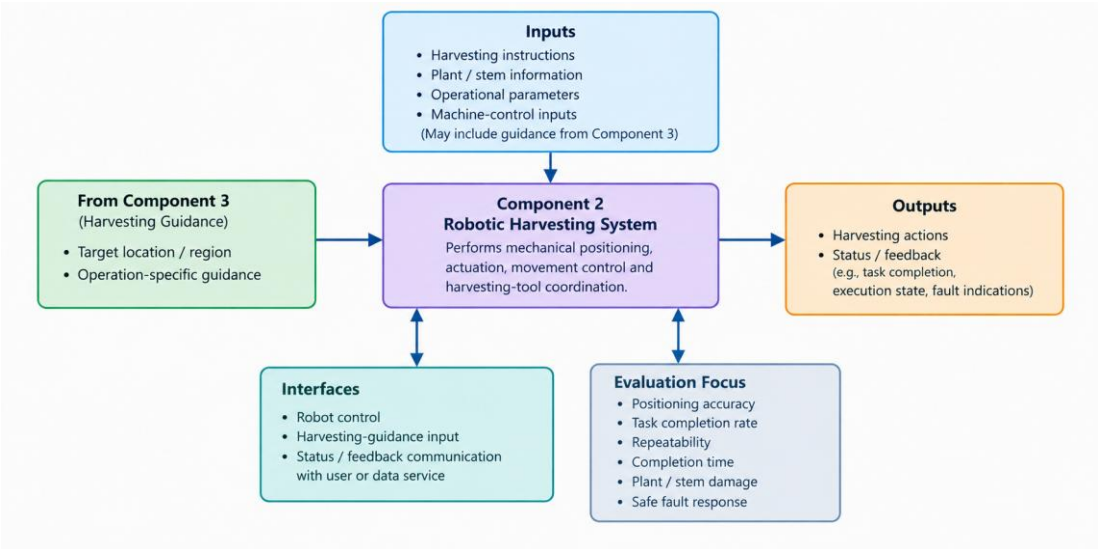


Figure 4 - component 2 diagram

Subproblem addressed: The physical and operational challenges of performing consistent cinnamon bark harvesting while controlling a mechanical harvesting system.

Inputs: Harvesting instructions, relevant plant or stem information, operational parameters, and machine-control inputs. Where implemented, Component 3 may provide a target or operation-specific guidance. Input formats and automation levels depend on the robotic design.

Core method: A robotic mechanism and control system perform supported harvesting actions through mechanical positioning, actuation, movement control, and harvesting-tool coordination. Control logic must translate permitted instructions into physical actions within defined operating limits.

Outputs: Physical harvesting actions and, where supported, status information such as task completion, execution state, or fault indications.

Contribution to the overall solution: Provides the physical execution capability that connects analytical guidance with harvesting. Component 3 supplies applicable guidance, while Component 2 determines how the mechanism can execute the supported action safely and consistently.

Main interfaces: Robot control, harvesting-guidance input, and status or feedback communication with the user or shared data service. These interfaces must be selected and tested during integration.

Evaluation focus: Positioning accuracy, successful task execution, repeatability, completion time, observed plant or stem damage, and safe responses to faults or invalid instructions. Criteria must reflect the prototype's actual capabilities.

Detailed mechanical design, actuator and controller selection, control algorithms, safety mechanisms, and experimental results are documented in Component 2's individual thesis.

3.5 Component 3 Summary: Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance

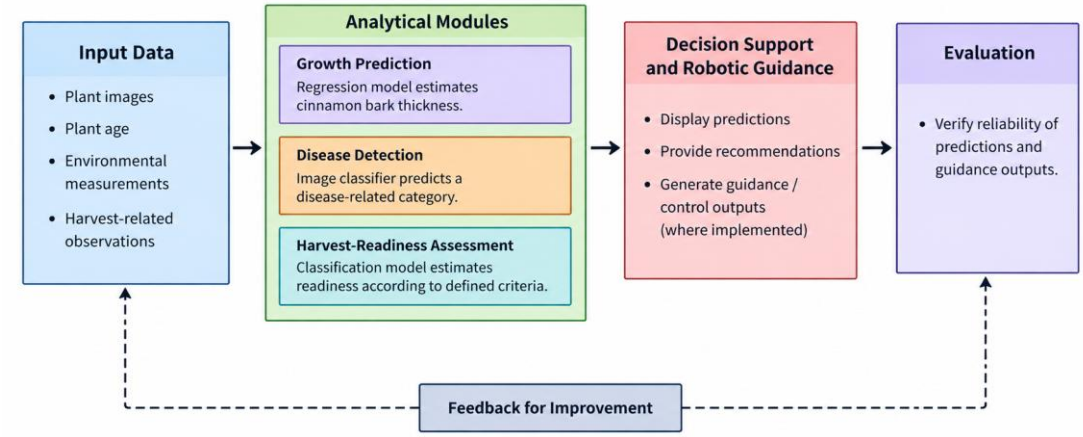


Figure 5 - component 3 diagram

Subproblem addressed: The difficulty of converting plant observations and historical or environmental data into consistent growth estimates, potential disease assessments, and useful harvesting guidance.

Inputs: Relevant Component 1 sensor measurements, historical observations, plant images where image-based analysis is implemented, and labelled datasets for model training or evaluation. Output reliability depends on input quality and availability.

Core method: Machine-learning methods support three distinct functions:

- **Growth prediction:** Uses suitable current or historical features to estimate a defined growth-related outcome.
- **Disease detection:** Analyses relevant observations or images to identify possible disease categories or symptoms.
- **Harvest guidance:** Combines available analytical results with defined decision rules to produce information supporting harvesting decisions.

These functions require separate evaluation because predicting growth, identifying potential disease, and recommending harvesting actions serve different purposes.

Outputs: Growth estimates, disease classifications or alerts, and structured harvesting recommendations or instructions. Where supported, outputs may include confidence scores, warnings, or indications that insufficient data prevent reliable predictions. Formats must be compatible with the monitoring and robotic interfaces.

Contribution to the overall solution: Provides analytical and decision-support capabilities by interpreting monitoring data and other observations. It connects analysis with physical harvesting by supplying guidance that Component 2 can interpret. Instructions must satisfy agreed operational and safety conditions before execution.

Main interfaces: Data input from Component 1, presentation of analytical results to users, and the guidance interface to Component 2. Model versions, input formats, output labels, units, and confidence thresholds must be documented for consistent interpretation.

Evaluation focus: Growth prediction using suitable errors between predicted and observed outcomes; disease detection using appropriate measures such as precision, recall, F1-score, and a confusion matrix; and harvesting guidance through recommendation correctness, consistency, usability, and correct interpretation by the robot. Final metrics must reflect the models, datasets, and intended conditions.

Detailed datasets, preprocessing, model architectures, training and validation methods, metrics, and experimental results are documented in Component 3's individual thesis.

3.6 Integration of the Three Components

Each component has a separate owner and technical responsibility, but the integrated solution depends on their cooperation through clearly defined interfaces.

Table 5 - Integration of the Three Components

Integration element	Responsible component or service	Purpose
Environmental data acquisition	Component 1	Collect relevant sensor measurements from the cultivation environment.
Monitoring and data availability	Component 1 and shared data service	Present readings and make suitable data available to other components.
Growth prediction and disease detection	Component 3	Analyse available observations and produce relevant estimates or classifications.
Harvesting guidance	Component 3	Convert applicable analytical results into structured recommendations or instructions.

Physical harvesting operation	Component 2	Execute supported harvesting actions within the robot's operating limits.
User interaction and oversight	Shared user interface	Allow users to review measurements, analytical outputs, and operational status.
Data exchange and history	Shared communication and data services	Maintain consistent formats and support information exchange between components.

The shared services represent functions required by the architecture, not proof that a separate server, cloud platform, or database has been implemented. They may operate on an embedded device, local computer, remote server, or combination of platforms.

Successful integration requires agreed data formats, units, timestamps, plant identifiers, communication protocols, and error-handling behaviour. Component 1 must supply interpretable data; Component 3 must produce documented outputs; and Component 2 must reject unsupported or invalid instructions and report relevant status where feedback is available.

Integration tests should address missing sensor readings, unavailable analytical services, communication failures, invalid instructions, and robot faults. When required inputs are unavailable, the system should report the limitation rather than present unsupported results as reliable. If machine-learning communication fails, the robot must remain in its independently defined safe operating state.

3.7 End-to-End User and Data Flow

The end-to-end workflow connects cultivation monitoring, analytical processing, harvesting guidance, and controlled robotic execution.

Step 1: Data collection. Component 1 acquires measurements from selected sensors, associating readings with their measurement types and available timestamps or plant/location identifiers. Basic validation identifies missing, invalid, or out-of-range readings.

Step 2: Data transfer and monitoring. Measurements are transmitted to the monitoring or data-management service. Users review available readings and changes over time. Where historical storage is implemented, previous measurements can be retrieved for comparison and analysis.

Step 3: Analytical processing. Relevant sensor measurements, historical observations, and available plant images are supplied to Component 3. Its models generate growth estimates and potential disease assessments where supported by the available data.

Step 4: Interpretation and harvest guidance. Analytical outputs are interpreted using defined decision rules to produce harvesting guidance where sufficient evidence exists. Guidance must be distinguishable from a confirmed harvesting command, and uncertain or incomplete results must be identified.

Step 5: User review and operational checks. The interface presents plant information and proposed harvesting actions. Depending on the design and risk controls, an authorized operator may approve an action. The system checks the instruction's validity, input availability, and the robot's permissible operating state.

Step 6: Robotic execution. Component 2 receives a valid instruction and performs the supported mechanical operation according to its control logic and operating limits. Safety controls must remain effective regardless of IoT or machine-learning service availability.

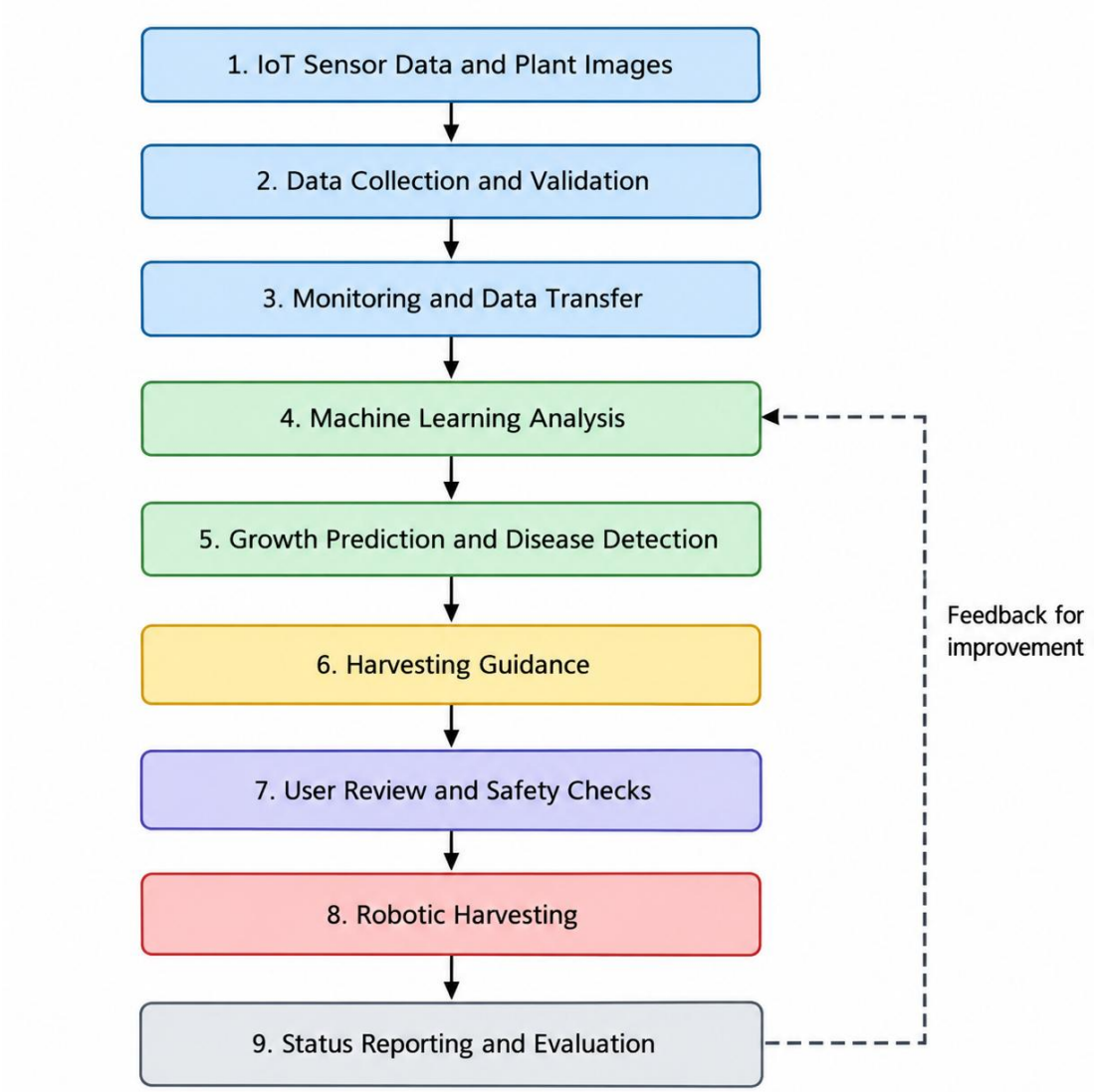


Figure 6 - user flow

Step 7: Status reporting and record keeping. Where feedback is implemented, robot status and task results are returned to the shared service for display. Records may include sensor observations, analytical results, harvesting guidance, user approval, and robot status, supporting evaluation and investigation of unsuccessful operations.

Step 8: Evaluation and improvement. Recorded observations and test results support integrated evaluation. Sensor readings can be compared with reference instruments, disease predictions with labelled observations, growth estimates with actual outcomes, and robotic performance through controlled trials. Retraining or updating models using collected records requires defined data-management and validation procedures.

This workflow connects monitoring, analysis, harvesting guidance, and controlled execution while distinguishing predictions from physical actions. Machine-learning outputs inform decisions but do not replace robotic safety requirements.

In summary, the proposed architecture integrates IoT monitoring, machine-learning analysis, and robotic cinnamon bark harvesting within a shared framework. The individual theses document the design and evaluation of each component, while the group report explains how their responsibilities and interfaces address the overall project problem.

4: COMPONENT INTEGRATION AND COHESIVE OPERATION

This chapter explains how the three components of the AI-Driven Smart Cinnamon Growth Monitoring and Automated Harvesting System work together. IoT monitoring collects environmental data, machine learning assesses plant growth, bark health, and harvest readiness, and robotics performs harvesting using validated target information.

Integration focuses on reliable data exchange, consistent interfaces, traceability, and safe coordination between software and hardware. Operator supervision and recorded outcomes support system evaluation and improvement.

4.1 Integration Strategy

A modular, incremental approach is used to simplify testing and identify failures.

1. IoT monitoring: Sensors collect temperature, humidity, soil moisture, and light intensity. Readings are time-tamped and linked to available plant or environment identifiers.
2. Machine-learning integration: Environmental data, plant images, and morphological measurements are analyzed to estimate growth, bark health, and harvest readiness.
3. Vision and robotic integration: Computer vision identifies the target bark region, and calibration converts its location into robot coordinates. Workspace and target validity are checked before harvesting.
4. Integrated decision-making: Plant readiness and target-localization reliability are evaluated before harvesting is allowed.
5. System evaluation: Component interfaces, end-to-end operation, and failure handling are tested.

Table 6 - Integration sequence

Stage	Main activity	Expected output
IoT monitoring	Collect and validate sensor data	Structured environmental records
Machine learning	Analyze data and estimate plant condition	Predictions and confidence values
Vision and robotics	Detect bark and validate target coordinates	Validated robot target
System integration	Connect interfaces and coordinate decisions	Integrated workflow

Evaluation	Test functionality and failure handling	Test results and performance evidence

Each component should be tested independently before integration. Actual hardware, communication protocols, and deployment arrangements must be documented according to the implementation.

The main risks include missing sensor data, communication failures, inconsistent identifiers, poor images, inaccurate calibration, and incorrect robot instructions. Input validation, explicit interfaces, supervised trials, and safety checks help reduce these risks.

4.2 Interfaces, Data Contracts and Shared Models

Interfaces define how sensor measurements, plant assessments, vision results, and robotic commands are exchanged. Shared data formats allow components to communicate consistently.

Plant and stem identifiers, timestamps, image references, and model versions help trace observations through predictions and harvesting outcomes.

4.2.1 IoT Sensor Data Interface

The IoT component supplies environmental readings to the monitoring and machine-learning components.

Table 7 - Environmental data fields

Field	Purpose
plant_id, stem_id	Identify the monitored plant or stem
environment_id	Identify the measurement location
timestamp	Record measurement time
temperature	Store temperature
relative_humidity	Store humidity
soil_moisture	Store soil moisture
light_intensity	Store light measurements
quality_flag	Identify invalid or questionable readings

Units, valid ranges, and quality rules must match the selected sensors. Missing or invalid measurements should be flagged rather than treated as reliable data.

4.2.2 Machine-Learning Data Interface

The machine-learning component receives validated sensor measurements, available plant morphology, and relevant images. It produces growth estimates, bark-health classifications, and harvest-readiness assessments.

Table 8 - Machine-learning inputs and outputs

Data	Purpose
Plant/stem ID and timestamp	Link predictions to observations
Environmental features	Provide cultivation context
Morphological measurements	Support growth and maturity assessment
Plant images	Provide visual evidence
Readiness output	Support harvesting decisions
Health output and confidence	Identify possible bark-health issues
Model version and quality status	Support traceability and reproducibility

Bark health and harvest readiness must be assessed separately. A healthy stem may not be ready for harvesting, while a mature stem may require further health assessment. Low-confidence predictions or missing inputs should trigger further observation or operator review.

4.2.3 Computer-Vision and Robotic Interface

The vision module identifies the target bark region and converts its image location into a physical robot coordinate using suitable calibration.

Table 9 - Vision-to-robot data fields

Field	Purpose
plant_id / stem_id	Identify the target
image_id	Identify the source image
target_region	Represent the detected bark area
localisation_confidence	Indicate detection reliability
calibration_id	Identify the coordinate transformation
target_pose	Provide the validated robot-frame target
workspace_status	Confirm whether the target is reachable
decision_status	Indicate approval or rejection

The controller must verify target freshness, calibration, confidence, and workspace limits. Missing coordinates or invalid calibration must prevent automatic harvesting.

4.2.4 Robot Feedback and Trial Records

Where implemented, robot records should include command identifiers, start and end times, execution status, errors, and safety events. Linking sensor observations, model predictions, target locations, and robot outcomes helps identify the causes of failed trials.

Interfaces may use REST APIs, JSON, databases, or other suitable mechanisms. The actual technologies and data formats must match the implemented system. Retries must not accidentally repeat a physical action.

4.3 Shared Nonfunctional Requirements

The integrated system must address reliability, performance, safety, data integrity, security, usability, and maintainability.

4.3.1 Reliability and Data Integrity

Sensors should identify invalid readings, machine-learning models should validate inputs, and the robot should reject unsafe commands. Consistent identifiers, timestamps, and error states help maintain data integrity. Stale or missing data must be distinguished from valid measurements.

4.3.2 Performance and Responsiveness

Sensor acquisition, data transmission, prediction, image processing, and robotic movement contribute to system latency. These should be measured separately to identify performance limitations. New sensor readings must not automatically trigger harvesting.

4.3.3 Safety and Controlled Robotic Operation

Harvesting requires both a suitable biological decision and a valid physical target. The robotic controller should:

- Check workspace boundaries and movement limits.
- Validate harvest-readiness and localisation-confidence thresholds.
- Recheck the target before contact where possible.
- Apply suitable speed and force limits.
- Provide emergency stopping and manual override.
- Respond safely to communication failures, localization loss, and abnormal motor conditions.

Failed safety checks must stop the harvesting sequence. Physical trials should be supervised, and autonomous operation should only be claimed when supported by completed tests.

4.3.4 Security and Privacy

Sensor data, plant images, model outputs, and robot records should be protected against unauthorized access or modification. Access controls, secure credential management, and appropriate backups should be used where applicable.

4.3.5 Usability and Accessibility

The dashboard should clearly present environmental readings, plant assessments, harvest readiness, warnings, and robot status. Predictions must be distinguished from measured values, and uncertain results should be explained where possible. Text labels should accompany colour indicators.

4.3.6 Maintainability, Scalability, and Reproducibility

Modular interfaces allow components to be updated independently. The system should maintain consistent identifiers as the number of plants, sensors, and images increases.

Reproducibility requires retaining relevant datasets, preprocessing configurations, model versions, calibration records, and trial logs. Training and test datasets should be separated appropriately to prevent information leakage.

Table 10 - Shared nonfunctional requirements

Requirement	Verification approach
Reliability	Component and fault-injection tests
Performance	Measure processing and harvesting times
Safety	Supervised safety tests
Data integrity	Validate schemas and identifiers
Security and privacy	Review access and configuration
Usability	Operator feedback and usability checks
Maintainability	Integration and regression tests
Reproducibility	Repeat evaluations using saved records

4.4 Integrated Workflow and Exception Handling

The integrated workflow connects environmental monitoring, machine-learning assessment, bark localisation, and robotic harvesting.

4.4.1 Normal Operating Workflow

The process follows these steps:

1. Observe: Collect environmental readings, plant information, and images.
2. Analyse: Estimate growth, bark health, and harvest readiness.
3. Localise: Detect the bark region and estimate its physical position.
4. Validate: Check prediction confidence, calibration, target freshness, and workspace limits.
5. Execute: Perform the approved harvesting operation within safety limits.
6. Record: Store execution status, duration, warnings, and outcomes.

If the plant is not ready or the target fails validation, harvesting must not proceed. Operator review or further observation may be required.

4.4.2 Exception Handling and Recovery

Failures may occur during sensing, prediction, localisation, communication, or harvesting.

Table 11 - Exception-handling strategy

Exception	Expected response
Missing or invalid sensor data	Flag the reading
Communication interruption	Report the failure and buffer data where supported
Missing plant identifier	Reject the record or request identification
Low prediction confidence	Request further observation or review
Stem not ready	Reject harvesting
Poor-quality image	Capture another image or request manual review
Invalid calibration	Block target execution
Target outside workspace	Reject movement
Communication loss during operation	Enter the configured safe state
Abnormal force or failed peeling	Stop and record the failure
Emergency stop	Interrupt automatic operation

Recovery depends on the failure type. Images may be recaptured, and temporary communication failures may permit data synchronisation. However, invalid calibration, unsafe movement, and emergency stops must never be bypassed by simply repeating a command.

If a fault occurs during movement, safe stopping takes priority over completing harvesting. All relevant states, command identifiers, and failure categories should be recorded.

4.4.3 Monitoring and Feedback

Recorded outcomes help evaluate each component. Environmental measurements can be compared with reference instruments, predictions with observed plant conditions, target coordinates with measured positions, and robotic results with harvesting success, damage, and cycle time.

A shared failure classification should distinguish prediction errors, localisation errors, calibration problems, mechanical failures, and safety interruptions. Evaluation must rely on recorded evidence rather than dashboard appearance alone.

4.5 Team Contribution and Dependency Matrix

The project assigns each member responsibility for one main component while sharing integration and evaluation tasks.

4.5.1 Individual Component Responsibilities

Member 1: IoT-Driven Smart Agriculture System

Responsible for sensor selection, environmental data acquisition, communication, data-quality checks, and preparation of structured records. Evidence includes sensor readings, acquisition logs, and communication tests.

Member 2: AI-Guided Robotic Cinnamon Bark Harvesting

Responsible for the robotic mechanism, motion control, target positioning, operational checks, and harvesting trials. Evidence may include positioning accuracy, task completion, cycle time, observed damage, and safety-test results.

Member 3: Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance

Responsible for dataset preparation, growth and readiness prediction, bark-health classification, model evaluation, and generating harvesting guidance. Evidence includes dataset descriptions, model versions, prediction logs, and evaluation results.

5: INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Introduction

This chapter describes the implementation and integration of the three project components: IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring, AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting, and Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance. Together, they form a cinnamon monitoring and harvest decision-support system that supports data-driven plant management and consistent harvesting decisions.

The system integrates a web interface, Python backend, machine-learning models, IoT data retrieval, cloud services, image processing, and a robotic control interface. It supports plant-health analysis, environmental monitoring, growth prediction, harvest-readiness assessment, and robotic harvesting functions. The backend handles requests, model inference, input validation, data retrieval, and communication with supported services.

The components share a common application of architecture while retaining distinct responsibilities. Environmental and plant-health data support growth and harvest assessments, whose recommendations inform the robotic harvesting workflow.

The system is an integrated software prototype. Implementing a feature or providing a control interface does not, by itself, demonstrate that the corresponding physical operation has been completed or validated under field conditions.

5.2 Common Technology Stack and Infrastructure

5.2.1 Frontend technologies

The frontend uses Next.js, React, and TypeScript. Next.js provides the application framework and routing, React supports reusable interactive components, and TypeScript improves consistency between interface components and backend data.

The application includes pages for disease detection, growth prediction, harvest readiness, IoT monitoring, robotic harvesting, history, and administration. Chart.js supports data visualization, while Axios handles HTTP communication in applicable modules. These interfaces present predictions, environmental readings, and operational messages.

5.2.2 Backend technologies

The backend uses Python and FastAPI to implement application programming interfaces (APIs), validate requests, process data, and return structured responses. Uvicorn runs the application as an ASGI server. The primary entry point is backend/main.py, and Python 3.11 is the recommended runtime.

Backend services handle model inference, IoT data retrieval, robotic-control functions, and supporting operations. This separation allows the frontend to request processing without implementing model inference or data preparation in the browser.

5.2.3 Machine-learning and image-processing technologies

TensorFlow supports the retained EfficientNetB0-based cinnamon leaf classification model, which processes prepared 224 × 224 RGB images and predicts one of seven leaf-condition classes. Results may include the predicted class, confidence, and related guidance.

Scikit-learn Random Forest models support bark-growth estimation and harvest-readiness assessment using relevant plant and environmental features. NumPy and pandas support numerical and structured data processing, while Pillow handles image decoding, orientation, resizing, and preparation.

These models support decision-making rather than replace field measurements or expert assessment. Their output must be interpreted according to their inputs, intended purposes, and available evaluation evidence.

5.2.4 Database, authentication, and external services

Firebase Authentication provides application authentication, while the Firebase Admin SDK supports backend token verification and Firestore access. Firestore stores supported application records.

Firebase Realtime Database (RTDB) supplies configured IoT telemetry. The backend retrieves records through HTTP requests and normalizes recognized fields for the monitoring interface.

Optional external services support supplementary disease verification and email notifications. Their operation depends on valid credentials, configuration, and service availability. Supplementary verification must not be treated as proven evidence of improved model accuracy.

5.2.5 Technology stack summary

Table 12 - Techstack

Technology	Main role in the system
Next.js 16.2.3	Frontend application framework and routing
React 19.2.4	Interactive user-interface components
TypeScript 5	Frontend typing and interface consistency
Python 3.11 (recommended)	Backend runtime
FastAPI 0.138.0	Backend API implementation
Uvicorn 0.49.0	Backend application server
TensorFlow	Leaf-image classification model
scikit-learn	Growth prediction and harvest-readiness models
NumPy and pandas	Numerical processing and structured data handling
Pillow 12.2.0	Image preparation and transformation
Firebase Authentication and Admin SDK	Authentication and backend identity verification
Firestore	Cloud persistence for supported application records
Firebase RTDB	Retrieval of configured IoT telemetry
Chart.js 4.5.1	Data visualisation
Axios 1.16.0	HTTP communication in applicable frontend modules
pyserial 3.5	Serial communication with the intended ESP32 interface
python-dotenv	Environment-based configuration

The versions shown above reflect the project dependency declarations reviewed during implementation documentation. The actual runtime behaviour remains dependent on installing compatible packages and supplying the required environment configuration.

5.3 Integrated System Architecture

5.3.1 Architectural overview

The architecture contains four logical layers:

1. **Presentation layer:** Web pages for uploading leaf images, viewing environmental readings, requesting growth and readiness estimates, and interacting with the robotic interface.
2. **Application-service layer:** FastAPI receives and validates requests, invokes processing functions, retrieves records, and returns structured responses.
3. **Intelligence layer:** Disease classification, growth prediction, harvest-readiness estimation, and related decision-support functions.
4. **Data and device integration layer:** Firebase services, RTDB telemetry retrieval, supported persistence, image-measurement interfaces, and the intended serial connection to the ESP32 relay controller.

Although these layers share an application, not every pathway constitutes a fully automated physical control loop. Some functions depend on external services, while certain robotic operations remain interface-level workflows or proposed mechanisms.

5.3.2 Data flows between components

The integrated system follows this sequence:

1. **Data acquisition:** Plant images, environmental readings, and plant-related values enter through the appropriate interface or configured source.
2. **Input preparation:** Images are decoded and resized, while numerical inputs are validated and converted to the expected format.
3. **Model processing:** The relevant model produces a disease classification, growth estimate, or readiness-related output.
4. **Decision support:** Predictions and configured checks help users interpret results.
5. **Robotic interaction:** The interface supports measurement and available relay-command functions. A readiness prediction does not prove that a harvesting command was issued.
6. **Record handling:** Supported records may be stored in configured cloud services or displayed through history interfaces.
7. **User review:** Users examine results and warnings before operational decisions.

This workflow combines the components while distinguishing predictions, recommendations, and physical execution.

5.4 Implementation of the Three Components

5.4.1 IoT-Driven Smart Agriculture System for Real-Time Cinnamon Plant Health Monitoring

The IoT component retrieves configured environmental data from Firebase RTDB through the backend endpoint /latest-iot-data/. The endpoint requests available records, interprets supported structures, and normalizes recognized sensor fields for the frontend.

The monitoring page polls for updates every five seconds and supports manual refresh. This is HTTP polling, not a WebSocket stream or direct browser subscription. The adapter recognizes variations in temperature, humidity, and soil-moisture field names and may return timestamps and battery information when available. Displayed readings depend on the source records and their quality.

A separate endpoint, /latest-sensor-data/, generates demonstration readings. These are not physical sensor measurements and must be labelled accordingly during evaluation.

The implementation provides software-based telemetry retrieval and display. Sensor accuracy, calibration, placement in cinnamon-growing environments, and long-term monitoring reliability require separate hardware and field evidence.

5.4.2 AI-Guided Robotic Machine for Automated Cinnamon Bark Harvesting

The robotic interface supports image-based measurement, cutting-depth estimation, and available relay-control interactions. It is intended to assist harvesting assessment and operator interaction with the proposed mechanism.

Image measurements depend on suitable calibration and a valid relationship between image and physical dimensions. Image coordinates cannot be used directly as robot coordinates without a validated transformation and physical calibration.

The backend endpoint /api/robotic-harvest/cut-depth estimates cutting depth using its implemented inputs and logic. Its existence does not establish validation against independent bark-thickness measurements or automatic physical blade control.

The application uses pyserial for the intended ESP32 relay interface and can dispatch supported commands while reporting remembered software states. These states do not independently confirm relay status, motor or blade position, or actual mechanical movement.

The harvesting workflow describes three mechanism stages:

- **First mechanism — roller or feed operation:** The interface provides relay-control interactions intended for feeding.

- **Second mechanism — longitudinal scoring:** The proposed mechanism uses a blade following the stem's curvature, but a complete physical cutter-control loop is not established by the reviewed implementation.
- **Third mechanism — slow-rotation peeling:** The proposed process coordinates movement and rotation, but a completed, validated physical peeling mechanism has not been established.

The scoring and peeling stages must therefore remain described as proposed or simulated unless physical construction and testing evidence is available. Likewise, the application's emergency-stop action cannot be claimed as a verified hardware safety mechanism without testing physical stopping behaviour and relevant failure conditions.

5.4.3 Machine Learning-Based Cinnamon Growth Prediction, Disease Detection, and Robotic Harvest Guidance

The ML component combines leaf-condition classification, growth estimation, and harvest-readiness assessment.

Disease detection. The retained EfficientNetB0 model processes prepared leaf images and predicts one of seven classes. Input dimensions, colour representation, and image quality affect the information supplied to the model. The predicted class is an indication of possible leaf condition, not a definitive plant-health diagnosis.

Growth prediction. A Random Forest regression model estimates bark-growth-related values from plant and environmental features. Its usefulness depends on representative training observations, valid feature definitions, and performance on independent data.

Harvest readiness. A Random Forest-based service uses inputs including plant age, rainfall-related information, phenology, and bark-browning information to generate readiness-related results and suggested actions. Interface-level checks and guidance must be distinguished from the model's own prediction.

Robotic harvest guidance. Growth and readiness outputs can inform an operator's decision to proceed. However, a displayed readiness result does not establish that a robot received a command or completed harvesting. Measurement, readiness assessment, and command execution must be validated separately before claiming an autonomous process.

Although the models are described as using real data, the available evidence does not establish verified numerical disease-classification accuracy, calibrated field

performance for growth prediction, or an independently validated harvest-readiness success probability. Such metrics should be reported only when supported by documented evaluation.

5.5 Development and Integration Process

5.5.1 Component-oriented development

Development was organised around the three components. IoT work focused on environmental retrieval and monitoring; robotics focused on image measurement, depth estimation, and relay interaction; and ML focused on disease classification, growth estimation, and readiness assessment.

This approach allowed each component to be developed around its own inputs and outputs. Integration required compatible data structures and clear boundaries between frontend operations, backend services, models, and external dependencies.

5.5.2 API-based integration

Development was organised around the three components. IoT work focused on environmental retrieval and monitoring; robotics focused on image measurement, depth estimation, and relay interaction; and ML focused on disease classification, growth estimation, and readiness assessment.

This approach allowed each component to be developed around its own inputs and outputs. Integration required compatible data structures and clear boundaries between frontend operations, backend services, models, and external dependencies.

5.5.2 API-based integration

The frontend and backend communicate through HTTP requests and structured responses. API routes provide entry points for disease prediction, growth estimation, readiness assessment, IoT retrieval, and robotic interaction.

Each endpoint requires appropriate validation: image-processing functions need valid image content, while prediction routes require correctly formatted numerical or categorical features. Authentication is also important for protected records and device commands.

The review identified a robotic integration discrepancy. The frontend references http://127.0.0.1:8000 for /api/robotic-harvesting/performance and /api/robotic-harvesting/analyze, but these endpoints were not found in the supplied backend. They cannot be considered integrated capabilities unless a separate implementation is verified.

Table 13 - Integration purpose

Interface or service	Integration purpose	Implementation consideration
Disease prediction service	Processes leaf images and return classification results	Input preparation and model output handling
Growth prediction service	Estimates growth-related values from plant and environmental features	Feature definitions and model input validation
Harvest-readiness service	Produces readiness-related predictions and guidance	Clear separation between model output and interface checks
/latest-iot-data/	Retrieves configured Firebase RTDB telemetry	External configuration, request failures, and record selection
/latest-sensor-data/	Provides demonstration sensor values	Must be labelled as simulated demonstration data
/api/robotic-harvest/cut-depth	Produces a numerical cutting-depth estimate	Requires independent measurement and physical validation
ESP32 serial interface	Dispatches supported relay commands	Serial availability and physical feedback limitations
Robotic image-analysis routes on port 8000	Referenced by the robotic frontend	Corresponding implementation was not identified in the supplied backend

5.5.3 Configuration and dependency management

The backend uses python-dotenv for supported environment settings. Firebase and optional external services require appropriate credentials and configuration. Frontend and backend dependencies are maintained separately to clarify package requirements and reduce compatibility issues.

Reproducible deployment documentation should specify runtime versions, packages, environment variables, model-artifact locations, database settings, device connections,

and service ports. Credentials must remain outside source code and public documentation.

5.5.4 Testing and defect management

Testing should include unit tests, API tests, integration tests, hardware tests, and separate field trials under real cinnamon-growing conditions.

The project report documents **29 passing selected unit tests** covering specific logic, schema validation, image preparation, provider-failure handling, metadata, and API compatibility. These results establish only that the selected checks passed, not that the complete test suite or integrated system was validated.

The review identified the following issues:

- Robotic image-analysis requests reference endpoints not found in the backend supplied.
- Harvest CSV persistence has a mismatch between declared headers and constructed row fields, potentially causing write failures.
- Harvest-history structures may not consistently match newer prediction-response fields.
- Relay status reports serial connectivity and remembered software states, not independently measured actuator states.
- External IoT retrieval depends on valid configuration and network availability.
- A complete physical feedback loop for automated cutting and peeling is not established.

These issues require integration and validation work. A successful API response does not prove that downstream operations completed successfully.

5.5.5 Model and data validation

Model evaluation must extend beyond successful execution. Disease classification should be assessed using a documented test set, class-wise performance, confusion matrices, and metrics appropriate to class imbalance. Growth predictions should be compared with independent measurements using suitable error metrics. Harvest-readiness predictions should be compared with independently recorded harvest or peeling outcomes.

Training and evaluation data must remain separate. Where appropriate, observations from the same plant or closely related samples should not be split across training and testing, as this could inflate reported performance.

Input definitions and units must also be consistent across components. Environmental features must match the growth model's expected definitions, and bark measurements used for cutting-depth estimation must be traceable to an appropriate measurement procedure.

5.6 Deployment and Operational Configuration

5.6.1 Application startup

The frontend and backend must be configured to communicate correctly. The backend is documented as running on port **8001**, whereas the robotic frontend references certain image-analysis routes on port **8000**. These service ports and responsibilities must be reconciled before the functions can be considered part of one operational service.

Deployment should include the following steps:

1. Install frontend and backend dependencies.
2. Configure backend environment variables and Firebase credentials.
3. Verify that required model artifacts exist at the expected locations.
4. Start the backend and test its API routes.
5. Start the front end and configure its backend URLs.
6. Verify authentication and configured Firestore and RTDB connections.
7. Check optional-service availability and report unavailable dependencies.
8. For robotic testing, verify serial-port configuration and the intended controller connection before sending commands.
9. Complete component and integration checks before operational use.

Each step must be verified and recorded for the deployment environment used.

5.6.2 Runtime monitoring and fault handling

The system depends on network connectivity, Firebase services, model artifacts, API processes, and serial communication, which can fail independently. It must distinguish valid predictions from failed requests, missing data, unavailable services, and unsuccessful persistence operations.

IoT readings should be interpreted using their source, timestamp, and retrieval status. A retrieved record may not represent the latest physical observation if records are not chronologically ordered. Missing or stale data should not be presented as confirmed current readings.

Invalid inference inputs or unavailable model artifacts must produce clear errors rather than misleading predictions. For robotics, command dispatch must remain distinct from confirmed physical movement: a missing serial response or unchanged software state does not establish the mechanism's actual state.

5.6.3 Data persistence and recovery

Configured Firebase services support persistence for applicable records, but individual writes can fail independently. In particular, the known harvest CSV field mismatch means a successful prediction does not guarantee a successful CSV write.

Deployment should verify each persistence operation, retain diagnostic errors, and define recovery procedures for failed writes. Backups and restoration procedures should cover required cloud records and model artifacts, while protecting credentials and configuration data.

Historical records should retain the source, timestamp, plant or observation reference, and relevant model or processing context to support traceability and investigation of differences between predictions and subsequent observations.

5.6.4 Deployment readiness

Table 14 - Deployment status

Area	Current implementation position	Additional requirement
Web interface	Pages are available for the main system functions	Full workflow and usability verification
Backend API	FastAPI entry point and multiple service routes are present	Complete route-by-route integration testing
Disease classification	Seven-class model artifact and inference path are present	Documented independent evaluation
Growth prediction	Random Forest regression functionality is present	Validation against independent measured growth values
Harvest readiness	Readiness model and related interface guidance are present	Validation against recorded real harvest outcomes
IoT monitoring	RTDB retrieval and periodic frontend polling are implemented	Sensor calibration and sustained field monitoring
Robotic image analysis	Frontend references image-analysis routes	Implement or verify the referenced service

Cutting-depth estimation	Numerical backend route is present	Independent measurement comparison and physical validation
Relay interaction	Serial dispatch and remembered command states are implemented	Verify hardware behavior and independent safety controls
Data persistence	Cloud persistence is supported; CSV writing has a known field mismatch	Correct field mapping and verify successful writes
Autonomous peeling	Later mechanism stages are described in the workflow	Construct, integrate, and validate physical mechanisms

The table distinguishes available software functions from readiness for physical field deployment. Field use requires the relevant software, data, integration, and safety checks to be completed.

5.7 Evidence of Team Execution

5.7.1 Coordinated development of the three components

The thesis combines environmental monitoring, ML-based plant assessment, and robotic harvesting support. IoT provides environmental observations; ML uses leaf images and plant or environmental features for condition, growth, and readiness outputs; and robotics provides measurement and supported device-interaction functions.

The objective of integration is to make these functions accessible through a common application. Compatible interfaces, consistent data definitions, error handling, and validation are therefore central to the project.

5.7.2 Evidence from implementation and testing

The project report documents the application structure, backend entry point, frontend pages, model artifacts, dependencies, API routes, IoT adapter, serial-command logic, and selected test results. These provide evidence for the functions present in the reviewed source.

The 29 passing selected unit tests demonstrate that software behaviors were checked. They do not establish that the full test suite passed or proved disease-classification accuracy, growth-prediction error, harvest success, cutting precision, or long-term sensor reliability.

A complete evidence package should include relevant source versions, model and data descriptions, test logs, interface screenshots, API test records, hardware-connection records where applicable, and independent evaluation results. Only evidence that exists should be reported as completed work.

6: INTEGRATED TESTING, EVALUATION AND VALIDATION

This chapter evaluates the integration of the three components of the AI-Driven Smart Cinnamon Growth Monitoring and Automated Harvesting System: IoT-based plant monitoring, machine-learning-based disease detection and growth prediction, and AI-guided robotic harvesting.

The evaluation considers software functionality, communication between components, model performance, user validation, and physical harvesting. Although selected software tests have passed, complete end-to-end validation and autonomous harvesting performance have not been established. Therefore, conclusions are based only on the available project evidence.

6.1 Evaluation Framework and Success Criteria

6.1.1 Evaluation Objectives

The evaluation focuses on whether:

1. The common application provides access to all three components.
2. The backend processes valid requests and returns appropriate results.
3. The IoT component retrieves and displays environmental data.
4. The machine-learning models provide disease classification, growth prediction, and harvest-readiness outputs.
5. The robotic interface supports its implemented measurement and command functions.
6. Data and results are transferred consistently between components.

Successful operation of one component does not necessarily prove that the complete system works correctly.

6.1.2 Evaluation Criteria

Table 15 - evaluation Criteria

Evaluation Area	Success Criterion
Application integration	Main functions communicate correctly
IoT monitoring	Environmental data are retrieved and displayed accurately
Disease detection	Valid images produce structured predictions
Growth prediction	Valid inputs produce interpretable estimates

Harvest readiness	Outputs support harvesting decisions
Robotic interaction	Commands are dispatched and failures reported
Data persistence	Records can be stored and retrieved consistently
Safety and reliability	Faults are handled without false completion claims
Usability	Users can understand outputs and complete tasks

These criteria define the required evaluation rather than confirming that every area has

6.2 Integration and End-to-End Testing

6.2.1 Integration Testing Approach

Integration testing examines communication between the frontend, FastAPI backend, machine-learning services, Firebase, and ESP32 relay interface.

The main integration pathways include:

- Frontend-to-backend communication for prediction services.
- Backend-to-Firebase communication for environmental data and persistence.
- Robotic measurement and cutting-depth estimation.
- Backend-to-ESP32 serial communication.
- Information flow between monitoring, prediction, and harvesting guidance.

Testing must verify both service responses and the correctness of transferred data.

6.2.2 API and Interface Testing

The backend uses FastAPI, while the frontend communicates through HTTP requests. The project reports 29 passing unit tests covering selected logic, schema validation, image preparation, provider-failure handling, metadata, and API compatibility.

However, these tests do not establish complete end-to-end validation. The source review identified missing backend implementations for the frontend's /api/robotic-harvesting/performance and /api/robotic-harvesting/analyze routes. These discrepancies must be corrected and retested to establish complete robotic image-analysis integration.

6.2.3 IoT Monitoring Tests

The IoT component retrieves environmental records through /latest-iot-data/ and periodically displays them in the frontend. Testing should verify data retrieval, temperature and humidity interpretation, soil-moisture values, timestamps, missing data, and connection failures.

The project also contains /latest-sensor-data/, which generates demonstration readings. These must be distinguished from actual sensor observations. Long-term sensor reliability, calibration accuracy, and field performance remain unverified.

6.2.4 Machine-Learning Integration Tests

The machine-learning component provides three functions:

- Disease detection: A seven-class classifier processes leaf images and returns structured predictions.
- Growth prediction: A Random Forest model estimates growth-related values from plant and environmental features.
- Harvest readiness: A model generates readiness-related outputs to support harvesting decisions.

Testing should verify input preparation, output formats, invalid-input handling, and consistency with the frontend. However, independent model-performance results are not fully documented. Functional execution should therefore not be confused with predictive accuracy.

6.2.5 Robotic Interface and Hardware Testing

The robotic component includes image-measurement functionality, cutting-depth estimation, and serial-command communication with the intended ESP32 relay interface.

Testing should cover command dispatch, communication failures, relay responses, stop commands, and emergency-stop behaviour. Software-maintained relay states only indicate commands tracked by the application; they do not confirm actual physical movement or blade position.

The available evidence supports the existence of a software control pathway, but complete physical scoring, peeling, and autonomous harvesting have not been demonstrated as fully implemented and validated.

6.2.6 Persistence and Regression Testing

The project supports cloud persistence, but a mismatch between the harvest CSV header and the fields generated by the persistence function may cause write failures. Differences between newer response fields and older history-page structures have also been identified.

These issues should be corrected, followed by regression testing to confirm consistency across prediction responses, CSV records, Firestore records, and history views.

6.2.7 Security, Performance, and Usability

Further testing is required in the following areas:

- Security: Authentication, authorization, input validation, and device-control access.
- Performance: Response times, error rates, and resource usage.
- Usability: Users' ability to interpret readings, predictions, and warnings.
- User acceptance: Structured feedback from representative users.
- Hardware safety: Physical stopping behavior and responses to faults.

The available evidence does not establish completion of a full security audit, formal performance benchmark, structured usability study, or independent hardware safety evaluation.

6.3 Overall Results

6.3.1 Summary of Available Evidence

The system integrates monitoring, prediction, and robotic-interface functions within a common frontend and FastAPI backend. However, some functions remain partially integrated or require additional validation.

Table 16 - Summary of Available Evidence

Evaluation Area	Result and Status
Common frontend	Main component pages are implemented
Backend services	Implemented, with integration checks required
Unit testing	29 selected tests passed
IoT monitoring	Retrieval pathway implemented; field reliability unverified
Disease detection	Seven-class model implemented; independent performance unverified

Growth prediction	Prediction function implemented; error measurements unavailable
Harvest readiness	Readiness function implemented; independent validation required
Cutting-depth estimation	Software function implemented; physical accuracy unverified
Robotic image analysis	Backend route discrepancies identified
Relay control	Command interface implemented; physical feedback unverified
Data persistence	CSV and history-field inconsistencies require correction
Autonomous harvesting	Full physical harvesting has not been demonstrated
Stakeholder validation	Formal evaluation evidence is unavailable

6.3.2 Interpretation of Results

The main achievement is the integration of monitoring, prediction, and robotic-interface functions into a common software environment. Environmental observations can support plant assessment, while machine-learning outputs can assist growth analysis and harvest-readiness decisions.

Nevertheless, software integration does not prove that a completely autonomous harvesting cycle operates successfully. Missing robotic routes, unverified physical feedback, and incomplete validation of scoring and peeling mechanisms remain important limitations.

Similarly, the existence of machine-learning models does not establish their accuracy on unseen plants or under different growing conditions. Independent testing is necessary before making such claims.

6.3.3 Component-Level Results

The individual theses should provide detailed evidence for each component:

- IoT component: Sensor specifications, calibration, environmental readings, and monitoring reliability.
- Robotic component: Measurement accuracy, cutting-depth evaluation, hardware communication, safety tests, and physical harvesting results where available.

- Machine-learning component: Real datasets, preprocessing, training methods, independent testing, and suitable performance metrics.

The common thesis should combine these findings only when the supporting evidence is documented and consistent.

6.4 Stakeholder and User Validation

6.4.1 Purpose

Stakeholder validation determines whether the system meets the needs of cinnamon growers, agricultural specialists, system operators, and technical reviewers. It can assess the usefulness of monitoring data, clarity of predictions, and practicality of harvesting guidance.

6.4.2 Validation Procedure

A structured evaluation could ask participants to:

1. Interpret Environmental Readings.
2. Submit a leaf image and review disease-classification results.
3. Interpret a growth estimate.
4. Review harvest-readiness guidance.
5. Examine robotic measurements or command status.
6. Identify warnings and situations requiring manual verification.

Feedback could be collected through questionnaires, interviews, and task observations. Physical harvesting claims would still require separate technical and safety tests.

6.4.3 Available Evidence

The project report documents implemented functions, selected tests, and integration issues. However, it does not establish a completed formal stakeholder study with documented participants, procedures, feedback, and analysis.

Therefore, no participant count, satisfaction score, or usability percentage can be reported without supporting records.

6.4.4 Remaining Concerns

Users must distinguish actual sensor readings from demonstration values, model predictions from verified plant conditions, and software commands from confirmed

physical actions. Future evaluation should determine whether users understand these distinctions and can operate the system safely.

6.5 Achievement of Common Objectives

Table 17 - Achievement of Common Objectives

Common Objective	Achievement Status
Develop a common platform	Partly achieved; integration gaps remain
Retrieve and display environmental data	Partly achieved; field reliability requires validation
Detect cinnamon leaf conditions	Partly achieved; independent model performance is unverified
Predict plant growth	Partly achieved; prediction error requires measurement
Assess harvest readiness	Partly achieved; comparison with actual outcomes is required
Support robotic measurement and depth estimation	Partly achieved; physical accuracy remains unverified
Provide robotic command functionality	Partly achieved; physical feedback and safety require testing
Maintain records and histories	Partly achieved; data inconsistencies require correction
Validate the integrated system	Partly achieved; full system validation remains outstanding
Demonstrate autonomous bark harvesting	Not demonstrated by the available evidence

6.5.1 Overall Objective Achievement

The project has established a common software platform that brings together monitoring, prediction, and robotic-interface functions. However, further evidence is needed to demonstrate reliable sensor operation, accurate predictions, consistent data storage, and safe physical control.

The objective of fully autonomous cinnamon-bark harvesting cannot be considered achieved based on the available implementation evidence.

6.6 Integrated Limitations and Threats

6.6.1 Data Quality and Representativeness

Model performance depends on the quality and representativeness of the real data used for development and evaluation. Differences in cinnamon varieties, plant age, growing conditions, imaging devices, and disease presentation may affect predictions. Independent testing and clear documentation of datasets and evaluation procedures are required.

6.6.2 Dependence Between Components

The system depends on correct information transfer between monitoring, prediction, guidance, and robotic interaction. Incorrect sensor readings or poorly calibrated measurements may lead to unsuitable decisions. Each data transfer must therefore be checked for accuracy, units, timing, and interpretation.

6.6.3 Software Integration and Persistence Risks

The identified robotic-route and CSV-field mismatches may interrupt workflows even when individual interface elements appear functional. These issues require correction and regression testing.

6.6.4 External Services and Deployment

IoT retrieval and cloud persistence depend on correct configuration and network connectivity. Differences between development and deployment environments may affect model loading, software compatibility, file paths, and device communication. Reproducible deployment procedures are therefore necessary.

6.6.5 Physical Robotics and Safety

The robotic interface does not establish a complete closed-loop harvesting system. Commanded relay states do not confirm physical movement, while scoring and peeling mechanisms remain unverified.

Physical testing must establish target identification, measurement calibration, suitable cutting depth, acceptable bark and stem damage, and safe stopping under normal and fault conditions. Human oversight and independently verified hardware safeguards remain essential.

6.6.6 Validity and Generalizability

The 29 selected unit tests cover specific software behaviours rather than the entire integrated system. In addition, the lack of complete independent model evaluations and formal stakeholder testing limits conclusions about predictive performance and usability.

Field conditions such as connectivity, lighting, sensor placement, plant variation, and mechanical constraints may also affect system performance. Future testing should use documented procedures, independent observations, repeated trials, and clear acceptance criteria.

6.6.7 Recommended Future Validation

The following activities are recommended:

1. Correct the robotic-route and persistence-field mismatches.
2. Complete API integration and regression testing.
3. Evaluate disease detection using an independent test dataset.
4. Compare growth predictions with independent measurements.
5. Validate harvest-readiness predictions against actual outcomes.
6. Verify IoT data sources, timestamps, calibration, and reliability.
7. Compare robotic measurements and cutting-depth estimates with reference measurements.
8. Test relay behavior, emergency stopping, and hardware failure responses.
9. Conduct a documented usability and stakeholder study.
10. Perform controlled end-to-end trials before claiming autonomous harvesting performance.

7: COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

This chapter presents a commercialization strategy for the integrated cinnamon monitoring and harvesting system. The recommended approach is phased implementation, beginning with prototype refinement and field testing, followed by pilot deployment and cooperative access. Commercial robotic harvesting should be introduced only after its safety, reliability, and economic benefits have been demonstrated.

The system combines IoT-based plant monitoring, machine-learning-based disease detection and growth prediction, and AI-guided robotic harvesting. Since it remains a prototype, commercial viability and full autonomous harvesting must be validated before the system is marketed as a complete operational solution.

7.1 Value Proposition and Beneficiaries

The system aims to combine environmental monitoring, plant-health assessment, growth prediction, harvest-readiness estimation, and harvesting guidance within a single platform. It could help cinnamon growers make informed decisions, improve monitoring efficiency, and support more consistent harvesting.

The main beneficiaries are:

Table 18 - main beneficiaries

Stakeholder	Potential Benefits
Smallholder farmers	Easier access to plant-health information and decision support
Plantation managers	Centralized monitoring, records, and improved planning
Cinnamon harvesters	Potential assistance with harvesting decisions and procedures
Farmer cooperatives	Shared equipment, training, and maintenance costs
Agricultural advisers	Structured observations and historical records
Technology partners	Opportunities for installation, integration, training, and support
Wider community	Potential improvements in resource management and agricultural productivity

These benefits remain proposed outcomes rather than proven commercial results. Pilot studies should measure prediction reliability, monitoring time, harvesting efficiency, plant damage, maintenance costs, and farmer satisfaction.

7.2 Market Alternatives and Differentiation

The initial target market includes cinnamon smallholders, farmer cooperatives, agricultural service providers, and commercial plantations in Sri Lanka.

Existing alternatives include manual inspections, conventional IoT devices, farm-management software, image-based disease detection, and traditional harvesting methods.

Table 19 - Market Alternatives

Alternative	Strengths	Limitations
Manual inspection	Familiar and flexible	Labor-intensive and potentially inconsistent
Conventional IoT monitoring	Provides environmental readings	May require additional analysis
Farm-management software	Supports planning and record keeping	May lack cinnamon-specific analysis
Image-based disease detection	Assists with symptom classification	Does not provide complete monitoring and harvesting support
Traditional harvesting	Established and experience-based	Relies substantially on manual labor
Proposed integrated system	Combines monitoring, prediction, and robotic guidance	Requires field validation, maintenance, training, and cost-benefit assessment

The main differentiator is the integration of three complementary functions within one cinnamon-focused platform. Potential long-term advantages include locally validated models, reliable hardware integration, relevant real-world datasets, and partnerships with grower organizations.

7.3 Business Model and Feasibility

7.3.1 Recommended Commercialization Model

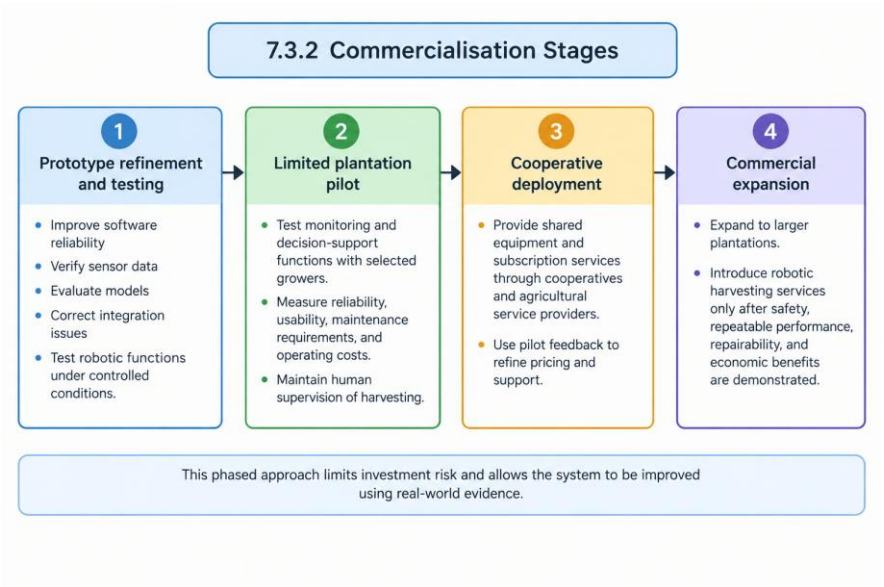
A hybrid model combining equipment installation, software subscriptions, and shared access through cooperatives is recommended. Farmers could initially use monitoring and decision-support services, with robotic harvesting introduced after appropriate validation.

Proposed revenue streams include:

- Installation fees: Sensor installation, configuration, and user onboarding.
- Software subscriptions: Dashboard access, historical records, disease analysis, and predictions.
- Equipment leasing or shared use: Access to equipment through cooperatives or agricultural service providers.
- Maintenance and support: Calibration, repairs, updates, and technical assistance.
- Training and integration: User training and adaptation to customer requirements.

These are proposed business options. Customer willingness to pay, final pricing, and recurring revenue have not yet been established.

7.3.2 Commercialization Stages



7.3.3 Resources, Partnerships, and Costs

Commercialization requires sensors, controllers, cameras, robotic components, computing infrastructure, software maintenance, real cinnamon plant data, and technical support.

Potential partners include farmers, cooperatives, agricultural research institutions, extension services, equipment suppliers, and software providers. These are proposed opportunities rather than confirmed agreements.

Table 20 - cost

Cost Category	Examples
Hardware	Sensors, controllers, cameras, robotic components, and replacements
Software and infrastructure	Hosting, databases, security, and maintenance
Development and validation	Engineering, data collection, and field testing
Deployment and support	Installation, training, travel, and repairs
Compliance and risk management	Safety assessments, documentation, and privacy controls

Financial feasibility should compare measurable benefits with operating costs.

\[\text{Annual Net Benefit}=\text{Annual Savings and Added Value}-\text{Annual Operating Costs} \]

The payback period can be estimated by dividing initial investment by annual net benefit when the net benefit is positive and sufficiently stable. Actual cost and performance data are required before any financial return can be claimed.

7.3.4 Technical and Operational Feasibility

The project provides a software foundation combining a web interface, backend services, Firebase, machine-learning models, and a robotic interface. However, commercial readiness requires reliable sensor readings, consistent data storage, validated model performance, and safe physical operation.

Frontend/backend discrepancies must be corrected, and demonstration readings must be distinguished from live sensor data. The system has potential technical feasibility

as a prototype, but operational and commercial feasibility remain to be established through field testing and cost-benefit analysis.

7.4 Adoption, Intellectual Property, and Compliance

7.4.1 Adoption Barriers and Mitigation

Potential barriers include equipment costs, connectivity limitations, digital skills, maintenance needs, plant-damage concerns, and uncertainty about prediction reliability.

Mitigation measures include:

- Providing a simple interface and practical user training.
- Designing equipment for plantation conditions.
- Explaining predictions and allowing users to report errors.
- Offering shared equipment and service-based access.
- Providing monitoring-only option before introducing robotic harvesting.

Pilot evaluations should examine usability, frequency of use, support requirements, willingness to pay, and the usefulness of recommendations.

7.4.2 Intellectual Property and Licensing

Potentially protectable elements include software, interface design, machine-learning implementations, system integration, and robotic workflows. Before commercialization, the team should establish ownership and licensing conditions.

The team should document individual contributions, dataset permissions, third-party software licenses, and any university or partner requirements. Version control and clear ownership agreements should be maintained.

Patentability, exclusive ownership, and freedom to operate must not be assumed without appropriate review.

7.4.3 Data Protection, Ethics, and Responsible AI

The system may process plant images, sensor readings, farm records, account information, and operational histories. Data linked to identifiable farmers or businesses may create privacy and confidentiality concerns.

Responsible practices include collecting only necessary data, restricting access, protecting credentials, securing stored and transmitted information where supported, and explaining how data may be used. Consent should be obtained where required, particularly when farm data are used to improve models or shared with other organizations.

Predictions should be treated as decision support rather than definitive conclusions. Uncertainty should be communicated where possible, expert review should remain available, and high-consequence actions should not depend solely on unverified predictions.

7.4.4 Safety and Regulatory Responsibilities

Robotic harvesting introduces risks from cutting mechanisms, moving parts, electrical equipment, and unexpected commands. Commercial deployment requires a documented safety assessment, physical guarding, emergency-stop mechanisms, safe operating procedures, and realistic hardware testing.

The system should enter a defined safe state following connectivity loss, sensor failure, invalid predictions, power interruptions, or communication errors. Safety controls must be verified on the physical hardware rather than inferred from dashboard messages.

Before deployment, the team must identify applicable requirements for machinery, electrical equipment, communications, worker safety, data protection, and business operations in the intended market. Compliance has not been established by the prototype description.

The team should document limitations honestly, avoid unsupported accuracy or autonomy claims, and ensure that users understand when human judgement is necessary.

7.5 Sustainable Development Goal Contribution

The system has potential to support several United Nations Sustainable Development Goals (SDGs). Its contribution depends on measurable benefits in real cultivation and harvesting conditions.

7.5.1 SDG 2: Zero Hunger

Relevant target: SDG 2.3 — Productivity and incomes of small-scale food producers

Although cinnamon is primarily a commercial spice crop, improved monitoring and harvesting decisions could support smallholder livelihoods. Potential benefits include earlier identification of plant-health concerns and better plantation planning.

These benefits should be assessed through monitoring efficiency, plant damage, harvesting productivity, and operating costs. Increased farmer income must not be claimed without field evidence.

7.5.2 SDG 9: Industry, Innovation and Infrastructure

Relevant target: SDG 9.5 — Enhancing scientific research and technological capability

The project integrates IoT, machine learning, web-based data management, and robotic assistance for cinnamon cultivation. Its contribution could be strengthened through reliable field trials, technical documentation, research collaboration, and maintainable equipment.

A prototype alone does not establish measurable improvements in industrial productivity or national research capacity.

7.5.3 SDG 12: Responsible Consumption and Production

Relevant target: SDG 12.2 — Sustainable management and efficient use of natural resources

Monitoring and harvesting guidance could support more efficient plantation management and reduce avoidable plant damage. Historical records could also help identify patterns in growing conditions.

However, inaccurate sensors, incorrect predictions, and unnecessary equipment use could waste resources. Sustainability benefits must therefore be assessed using actual farm records.

7.5.4 Potential Negative Effects and Mitigation

Table 21 - Potential Negative Effects and Mitigation

Potential Risk	Consequence	Mitigation
Incorrect predictions	Poor treatment or harvesting decisions	Field validation, uncertainty communication, and expert review

Robotic failure	Injury or plant damage	Physical safeguards, supervised operation, and fault testing
High costs	Limited access for smallholders	Cooperative ownership, leasing, and affordable monitoring packages
Electronic waste	Increased resource consumption	Repairable components and responsible disposal
Data misuse	Privacy or commercial confidentiality concerns	Access controls, secure storage, and transparent policies
Overdependence on automation	Inappropriate decisions	Human oversight, training, manual override, and safe fallback procedures

7.5.5 Measuring and Sustaining SDG Benefits

Pilot deployments should measure:

- Time required for plant monitoring.
- Agreement between disease predictions and expert assessments.
- Agreement between harvest-readiness predictions and observed plant conditions.
- Harvesting errors, bark damage, and equipment faults.
- Operating and maintenance costs.
- Farmer satisfaction, accessibility, and willingness to continue.
- Equipment service life, repair rates, and disposal practices.

Baseline measurements should be collected before deployment and compared over a suitable period. Where possible, trials should include multiple farms and account for differences in plantation size, cultivation practices, and environmental conditions.

8: CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The common problem addressed by this project is the need for a more integrated and data-informed approach to cinnamon plant health monitoring, growth assessment, disease detection, and harvesting. Conventional plantation management relies substantially on manual observation, practical experience, and labor-intensive harvesting activities. Although these methods remain important, they present opportunities for improvement through digital monitoring, machine-learning-based decision-making support, and appropriately designed automation.

To address this problem, the team developed an integrated prototype combining three complementary components: an IoT-driven smart agriculture system for real-time cinnamon plant health monitoring, an AI-guided robotic machine for automated cinnamon bark harvesting, and a machine-learning-based system for cinnamon growth prediction, disease detection, and robotic harvest guidance. Together, these components establish a common technological approach that connects plantation observations, data analysis, and harvesting-related decisions.

The project demonstrates the feasibility of bringing these functions together within software and hardware-oriented architecture. The system includes a web-based interface, backend services, Firebase-based data management, machine-learning components for plant-health classification and growth-related predictions, and a robotic harvesting workflow intended to support physical operations. The integrated design provides a foundation for collecting and presenting agricultural information and connecting analytical outputs with harvesting assistance.

However, the evidence available from implementation and selected testing does not establish that the complete system is ready for unrestricted field deployment or fully autonomous harvesting. The existence of individual software components does not, by itself, demonstrate reliable end-to-end operation under real plantation conditions. Further verification is required for model performance, live sensor integration, physical robotic control, operational safety, and sustained performance over time. Commercial benefits, including reduced labor costs, improved harvesting efficiency, and increased farmer income, also remain to be demonstrated through structured field evaluation.

Accordingly, the project addresses the common problem by establishing an integrated prototype and a practical foundation for further development, rather than by claiming that all challenges associated with cinnamon monitoring and harvesting have been fully resolved. Its principal outcome is the demonstration of how IoT, machine

learning, and robotic assistance can be organized into a common cinnamon-focused system. The next stage must establish whether this integration delivers measurable, reliable, safe, and economically meaningful benefits to growers and harvesters.

8.2 Principal integrated contribution

The principal contribution of the project is the integration of plant monitoring, machine-learning-based agricultural decision support, and robotic harvesting assistance into a unified cinnamon-focused system. Rather than treating plant monitoring, disease detection, growth prediction, and harvesting as isolated activities, the project explores how information from these functions can contribute to a connected agricultural workflow.

The integrated contribution can be understood through four main outcomes.

First, the project establishes a shared digital platform for cinnamon management. The web interface and backend architecture provide a common environment through which users can access monitoring information, plant-health analysis, growth-related predictions, harvesting guidance, and associated records. This provides a basis for connecting observations and decisions that might otherwise be managed separately.

Second, the project connects agricultural data with decision support. The IoT component is intended to provide monitoring information, while machine-learning components analyses plant images and relevant growth or harvesting information. The integration illustrates how different types of agricultural data can be used to support decisions. The reliability and practical usefulness of those decisions must still be assessed against real-world observations.

Third, the project establishes a pathway from digital analysis towards physical harvesting assistance. The robotic component introduces a workflow in which harvesting-related guidance can be linked to a proposed physical operation. This is an important integration challenge because a prediction generated by software cannot be transferred directly to a machine without suitable target localization, calibration, control verification, and safety checks. The project therefore highlights the importance of coordinating software intelligence with the constraints of physical agricultural work.

Fourth, the project provides a foundation for continued research and practical evaluation. The combined architecture creates opportunities to investigate model reliability, sensor integration, harvesting precision, user acceptance, and commercial feasibility within a single project framework. It also identifies the need to distinguish

implemented functions from simulated operations, externally dependent services, and capabilities that have not yet been validated on physical hardware.

Beyond the three individual components, the team's broader contribution is the development of a common system concept, an integrated software architecture, and a set of implementation and evaluation requirements for cinnamon-focused smart agriculture. The project also demonstrates that successful integration requires more than combining technologies: it requires consistent interfaces, reliable data handling, explicit safety conditions, clear responsibility for each subsystem, and evidence that the complete workflow behaves as intended.

The value of this contribution lies both in the prototype and in the engineering knowledge gained through integrating distinct agricultural technologies. It provides a starting point for future researchers and developers who wish to move from separate monitoring and prediction tools towards a more coordinated system for cinnamon cultivation and harvesting.

8.3 Future directions

Future development should follow a staged approach, prioritizing reliability, evidence, and safety before expanding the system's autonomy or commercial scope. The following directions are recommended in order of practical importance.

8.3.1 Strengthen integrated system reliability

The first priority is to resolve the technical issues identified during implementation and integration testing. These include verifying that frontend requests correspond to available backend endpoints, ensuring consistent communication between services, correcting data-persistence issues, and confirming that historical records are stored and retrieved as intended.

The system should also distinguish live IoT telemetry from simulated sensor readings. Each data source should be clearly identified so that users and evaluators can determine whether information originates from physical sensors, a test environment, or another service.

Integration testing should be extended beyond selected unit tests to cover complete workflows, including data collection, backend processing, display of results, persistence, error handling, and recovery after communication failures. These improvements are necessary to establish a reliable foundation for field testing.

8.3.2 Validate machine-learning models using real field observations

The disease-detection, growth-prediction, and harvest-readiness models should be evaluated systematically using real cinnamon plant observations and clearly documented reference labels. The project uses real data, but this alone does not establish the generalization performance of the models. Their reliability must be demonstrated through appropriate evaluation.

Future work should include:

- Documenting the origin, collection conditions, labelling process, and characteristics of the datasets.
- Separating training data from validation and test data to reduce the risk of data leakage.
- Evaluating disease classification with metrics such as precision, recall, F1-score, and a confusion matrix.
- Comparing growth predictions with observed measurements using suitable regression metrics.
- Evaluating harvest-readiness predictions against expert assessments and actual harvesting decisions.
- Testing performance across different plants, locations, lighting conditions, growth stages, and environmental conditions.
- Recording uncertain or incorrect predictions and determining how they should be presented to users.

Where possible, evaluation should include independent field observations not used during model development. This would provide stronger evidence of performance beyond the original development dataset.

8.3.3 Improve IoT monitoring and field deployment

The IoT subsystem should be tested using physical sensors deployed in representative cinnamon-growing environments. Testing should establish whether the sensors produce consistent readings, how frequently measurements can be collected, and how the system behaves when connectivity or power is interrupted.

Calibration and maintenance procedures should be documented, and sensor readings should be checked against appropriate reference instruments where applicable. The platform should also record measurement timestamps, device identifiers, and data-

quality indicators to make it easier to identify missing, delayed, or implausible readings.

Future deployments should assess the practicality of different connectivity arrangements and the system's ability to operate when internet access is intermittent. These improvements would help establish whether the monitoring platform can support routine plantation use rather than only controlled demonstrations.

8.3.4 Develop and validate robotic harvesting safety

Robotic harvesting requires particular attention because incorrect physical actions can damage plants or endanger operators. Before attempting fully autonomous harvesting, the system should be tested incrementally, starting with controlled movements and supervised operation.

Future development should prioritise:

- Accurate calibration between camera coordinates, detected targets, and robot coordinates.
- Verification of target localisation and cutting-depth estimates against physical measurements.
- Safe control of actuators, including testing of emergency-stop functions and fault responses.
- Defined safe states for communication loss, invalid predictions, sensor failure, and unexpected movement.
- Mechanical protection, operator access controls, and documented operating procedures.
- Supervised trials that measure harvesting accuracy, bark damage, cycle time, repeatability, and equipment reliability.

The robotic system should not be treated as fully autonomous until the physical operation has been validated repeatedly under representative conditions. Human oversight should remain in place during development and early deployment.

8.3.5 Conduct a structured field pilot and stakeholder evaluation

Once the principal integration and safety issues have been addressed, the next stage should be a pilot involving cinnamon growers, harvesters, and relevant agricultural advisers. The pilot should examine both technical performance and the practical experience of using the system.

A suitable evaluation would establish baseline conditions before deployment and then compare them with results obtained when the system is used. Measurements could include monitoring time, the agreement between model predictions and expert assessments, harvesting time, plant damage, maintenance frequency, and user-reported usefulness.

The evaluation should include clear research questions, documented procedures, appropriate participant consent, and transparent reporting of limitations. The number and characteristics of participating farms should be determined according to the intended study and available resources, rather than assumed in advance.

Feedback should inform improvements to the interface, alert design, prediction explanations, training materials, and operating procedures. This would help establish whether the system addresses actual user needs and whether its benefits justify the effort and cost of adoption.

8.3.6 Establish commercial and operational feasibility

Commercial development should proceed only after the pilot provides sufficient evidence of technical reliability and user value. Future work should compare the total cost of the proposed system with the cost and performance of existing plantation management and harvesting practices.

The assessment should consider equipment purchase or leasing, installation, connectivity, software services, maintenance, repairs, training, and replacement of components. It should also assess potential benefits using measured evidence rather than assuming that automation necessarily reduces costs.

A phased commercialisation strategy is appropriate. Monitoring and decision-support functions could be offered first, followed by shared robotic harvesting services after physical safety and harvesting performance have been demonstrated. Cooperative access models may help make the technology more accessible to smallholder growers who cannot justify purchasing a complete system independently.

8.3.7 Improve responsible data management and sustainability

Future versions should strengthen data governance, user access controls, documentation of data sources, and transparency about how information is used. Dataset permissions and third-party software licences should be reviewed before wider distribution or commercial deployment.

The project should also consider equipment durability, repairability, energy consumption, and electronic waste. Where practical, components should be replaceable and maintainable so that system upgrades do not require unnecessary disposal of functioning equipment.

Sustainability should be assessed through measurable outcomes, including changes in monitoring effort, equipment lifetime, resource use, plant damage, and operating costs. These indicators would help determine whether the system produces benefits that are environmentally and economically meaningful.

8.3.8 Extend the system through continued research

After the core system has been validated, additional research could investigate more advanced growth forecasting, improved disease recognition, multimodal analysis combining sensor and image information, and more adaptive harvesting guidance. The integration of additional data sources may improve decision-making, but each extension should be justified by a clear research objective and evaluated against an appropriate baseline.

Longitudinal observations could also support investigation of changes in plant growth, disease occurrence, and harvesting readiness over time. Any future model should be tested for reliability across different growing conditions before its outputs are used to trigger consequential physical actions.

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